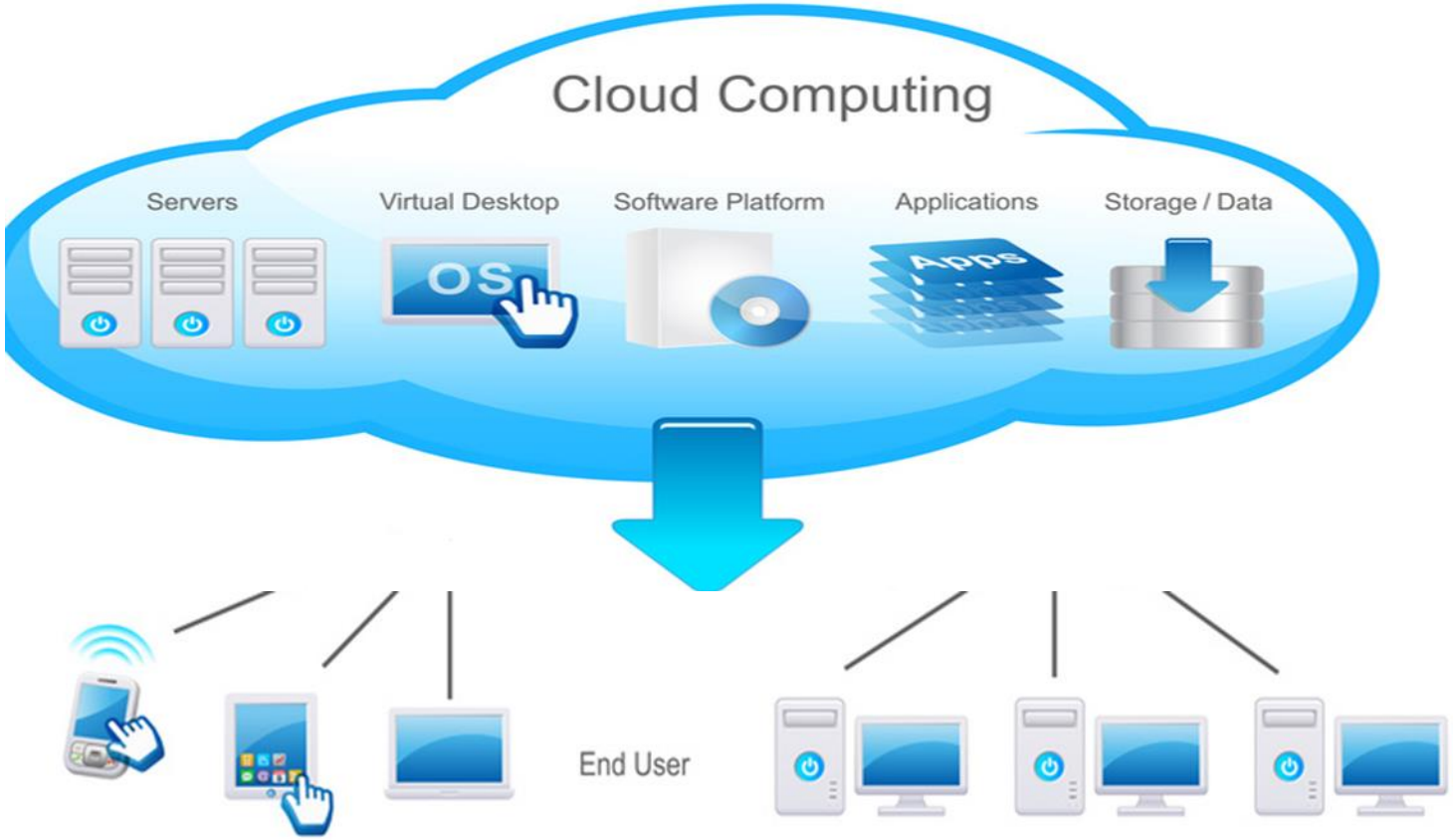
# CLOUD COMPUTING

- Cloud computing is a **service provisioning technique** where **computing resources** like **hardware such as servers and storage devices**, **software's** and **complete platform for developing applications** are provided **as a service by the cloud providers to the customers.**



# CLOUD COMPUTING (Cont…)

- Customers **can use these resources as and when needed**, **can increase or decrease resource capacities dynamically** according to their requirements and **pay only for how much the resource were used**.

- Customers **no need to invest money to purchase**, **manage** and **scale infrastructures**, **software upgradation** and **software licensing.**

# Cloud Service Models

- The services that are provided by the cloud providers are broadly classified into three categories:
  - **Infrastructure-as-a-Service (IaaS)**
  - **Platform-as-a-Service (PaaS)**
  - **Software-as-a-Service (SaaS)**

# Cloud Service Models

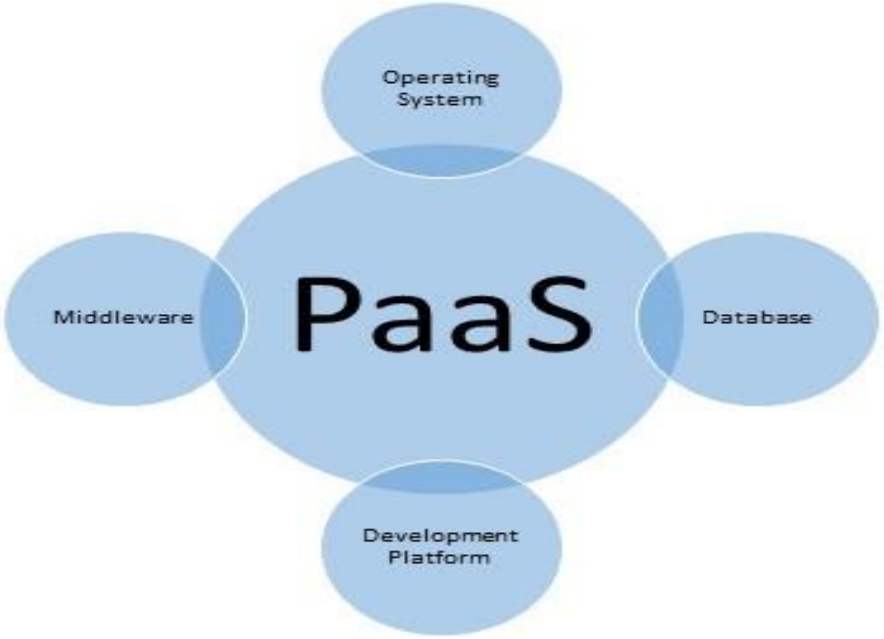
- **Infrastructure-as-a-Service (IaaS):** In Infrastructure-as-a-Service model, the service provider owns the hardware equipment's such as **Servers, Storage**, **Network** and is **provided as services** to the clients. The **client uses these equipment's and pays on per-use basis.**

- E.g. **Amazon Elastic Compute** (EC2) and **Simple Storage Service** (S3).

# Cloud Service Models

- **Platform-as-a-Service (PaaS):** In Platform-as-a-Service model, **complete resources** needed to **Design**, **Develop**, **Testing**, **Deploy** and **Hosting** an application are provided as services **without spending money for purchasing and maintaining the servers, storage and software**.

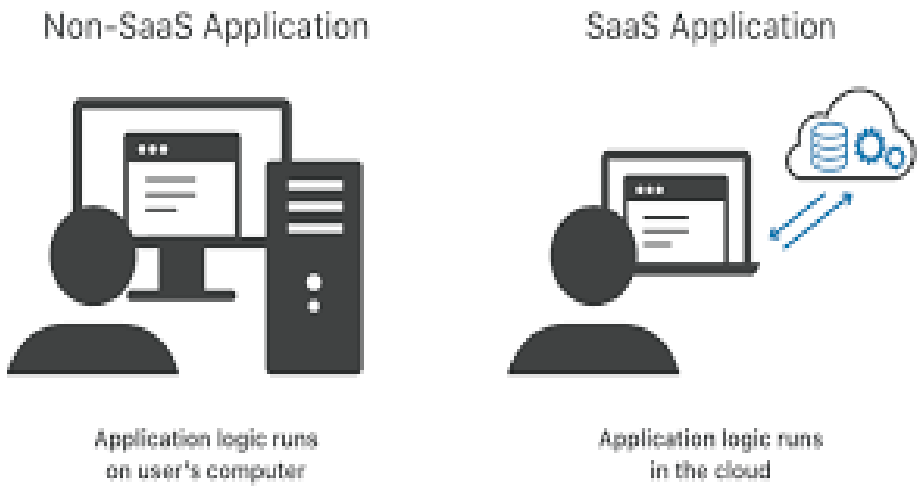
- **PaaS is an extension of IaaS**. **In addition to the fundamental computing resource supplied by the hardware** in an IaaS offering, **PaaS models also include the software and configuration required to create an applications**.




- E.g. **Google App Engine**.

# Cloud Service Models

- **Software-as-a-Service (SaaS):** In Software-as-a-Service model, the service provider provides **software's** as a service over the Internet, eliminating the need to **buy**, **install**, **maintain**, **upgradation and licensing** on their local machine.




- E.g. **Accounting, CRM, Google Docs** are all popular examples of SaaS.

# Cloud Deployment Models

- Mainly there are four cloud deployment models ( 4 ways we can create/organize a cloud)
  - **Public Cloud**
  - **Private Cloud**
  - **Community Cloud**
  - **Hybrid Cloud**

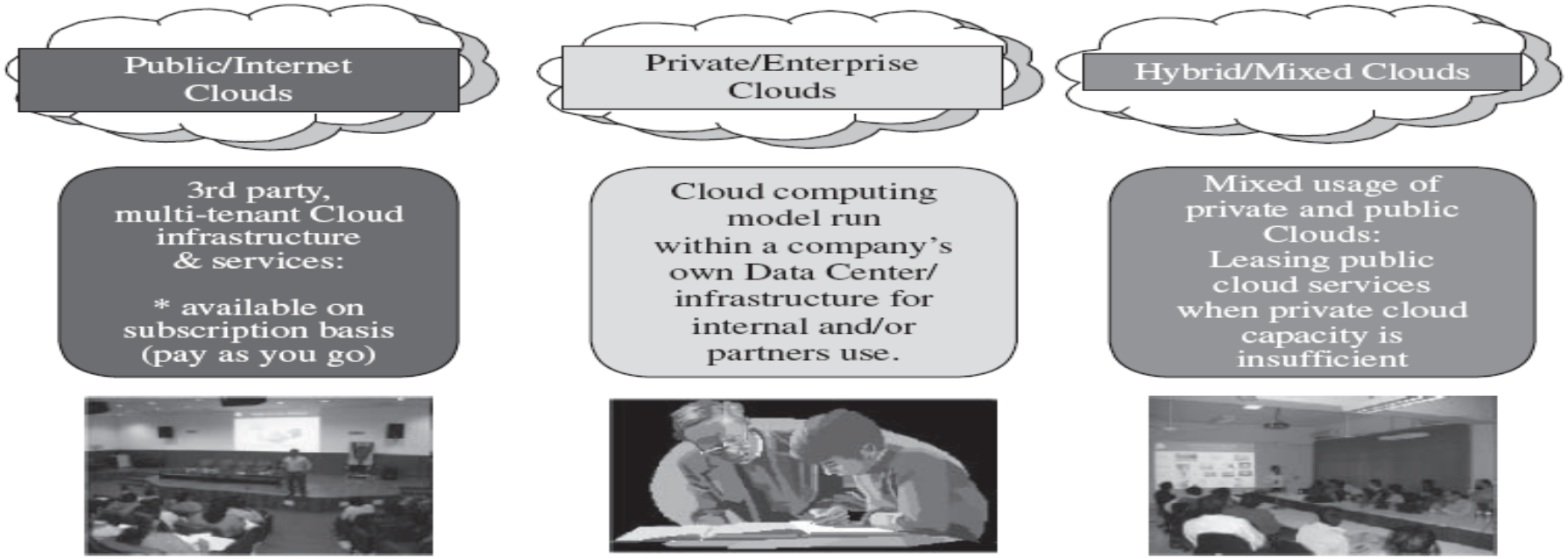



FIGURE 1.4. Types of clouds based on deployment models.

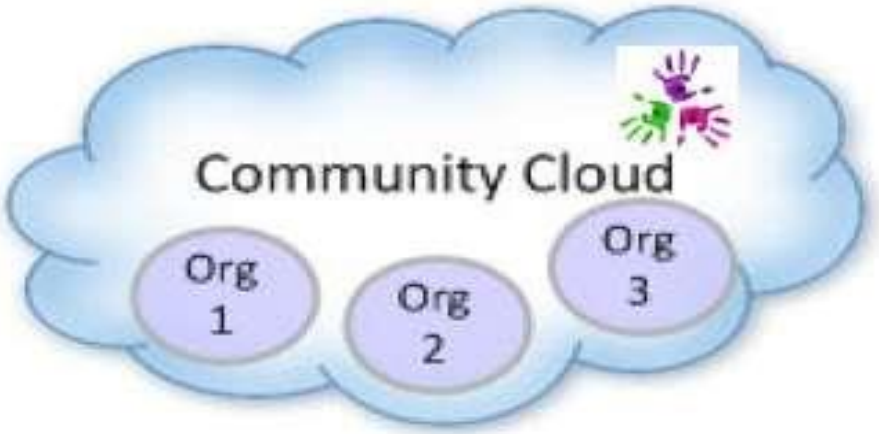
# Cloud Deployment Models (Cont…)

- **Public Cloud**: A public cloud is a cloud in which **services and infrastructure are hosted off-site by a cloud provider** (owned by an organization selling cloud services) and easily **accessible to general public via internet**.

- **Private Cloud:** Private Cloud is a cloud where **services and infrastructure are operated for a single operation accessible via private network**, managed internally or by a third party. It is greater level of security.

# Cloud Deployment Models (Cont…)

- **Community Cloud:** Community Cloud is a cloud where **services and infrastructure are accessible by a group of organizations**.




- **Hybrid Cloud:** Hybrid Cloud is a cloud which is a **mixture of private and public cloud**. In this type of cloud **all critical and sensitive applications and data are stored in private cloud** and **non critical and non sensitive applications and data are stored in public cloud**.

# Features of Cloud Computing

- **It is elastic:** Cloud computing is flexible in nature, where users can **scale up** and **scale down** the resources as needed.

- **Pay per use:** Usage is metered and user **pays only for how much the resources were used**.

- **Operation:** The services are completely **handled by the provider**.

- **Reduce capital cost:** No need to **invest money** on purchasing and maintaining of hardware and software, software licensing, training required for IT staff.

- **Remote accessibility:** Users can access **applications and data stored on cloud** from anywhere any time worldwide through a device with internet connection.

- **Better use of IT staff:** Staff with in enterprise need not worry on purchasing and maintaining of servers, softwares, up gradation of servers and softwares, software licensing etc., instead they can concentrate more on work.

# Cloud Services Examples:
## IaaS-Amazon EC2, Google Compute Engine, Azure VMs

# Amazon EC2

- **Amazon Elastic Compute Cloud** is an Infrastructure as a Service offering from Amazon.
- EC2 is a web service that **provides a computing capacity in the form of virtual machines**.
- Amazon EC2 allows **users to launch instances on demand using a simple web based interface**.
- Amazon provides **pre-configured Amazon Machine Images** (AMIs) which are templates of cloud instances.

| | Small | Large | Extra Large | High CPU-Medium |
|---|---|---|---|---|
| Compute unit | 1 | 4 | 8 | 5 |
| Memory | 1.7 GB | 7.5 GB | 15 GB | 1.7 GB |
| Storage | 160 GB | 850 GB | 1690 GB | 350 GB |
| Platform | 32 bit | 64 bit | 64 bit | 32 bit |

- Users can also **create their own AMIs with custom applications, libraries and data**.

# Cloud Services Examples:
## IaaS-Amazon EC2, Google Compute Engine, Azure VMs

## Amazon EC2

- Amazon EC2 also provides **instances with high memory, high CPU resources, Cluster Compute instances, Cluster Graphical Processor unit (GPU)instances and high Input/Output instances**.

- Instances can be launched with a **variety of operating systems**.

- Users can **load their applications on running instances** and rapidly and easily **increase or decrease capacity to meet dynamic application performance requirements**.

- With EC2, **users can even provision hundreds or thousands of server instances simultaneously**, manage network access permissions and monitor usage resources through web interface. (Create 2 VMs at 3 different places and running applications, creating network among 3 and allowing data transfer among 3 VMs and monitoring resource usage)

# Cloud Services Examples:
## IaaS-Amazon EC2, Google Compute Engine, Azure VMs

### Amazon EC2

- Amazon EC2 provides **instances of various computing capacities ranging from Small Instances** (Eg: 1 Virtual core with 1 EC2 compute unit, 1.7 GB memory and 160 GB instance storage) to **Extra Large Instances** (Eg: 4 Virtual cores with 2 EC@ compute unit each with 15GB memory and 1690 GB instance storage).

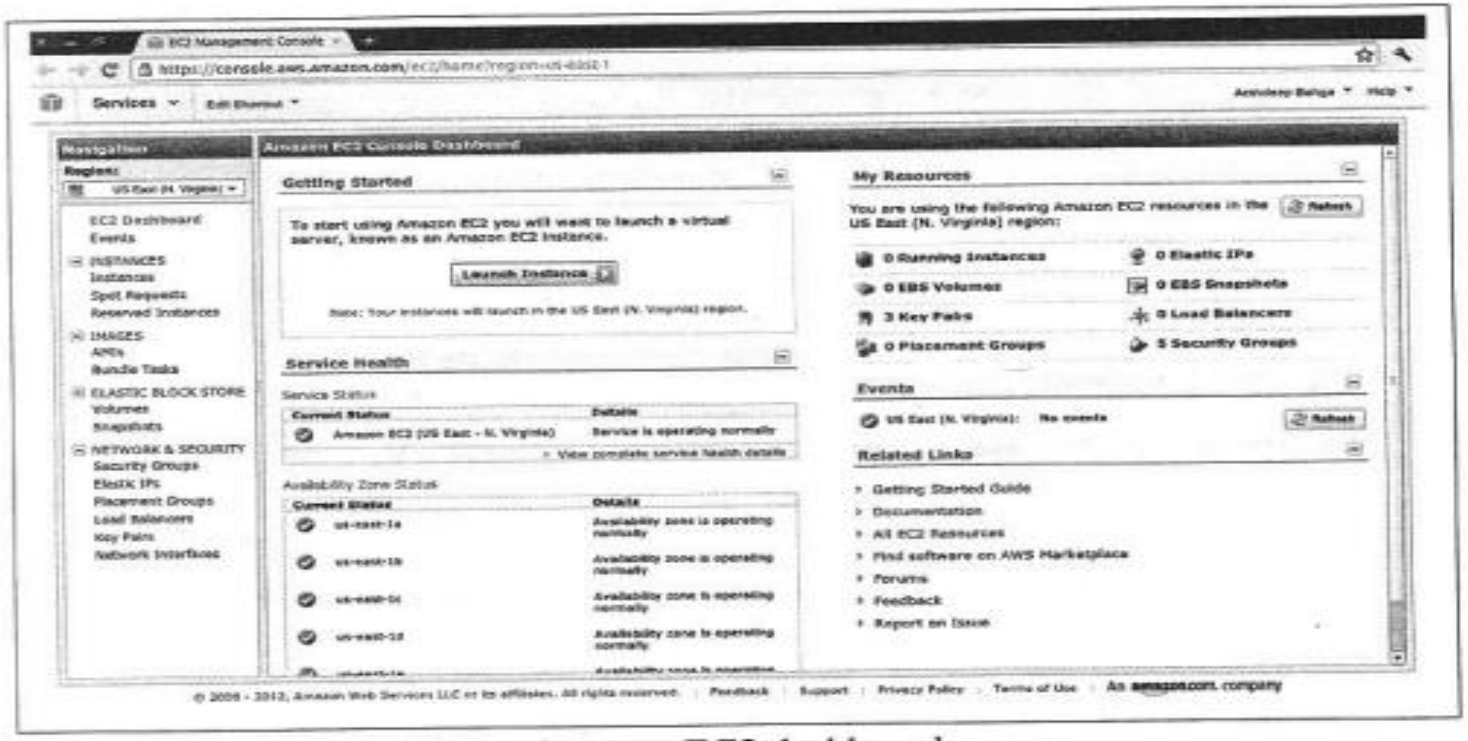
- The **pricing model for EC2 instances is based on Pay-Per Use model**. Users are billed based on the number on instance hours used for on demand instances.

- EC2 also provides **spot instances that allow users to bid on unused Amazon EC2 capacity and run those instances for as long as their bid exceeds the current spot price**.

# Cloud Services Examples:
## IaaS-Amazon EC2, Google Compute Engine, Azure VMs

## Amazon EC2

- The below figure shows screenshot of Amazon EC2 dashboard



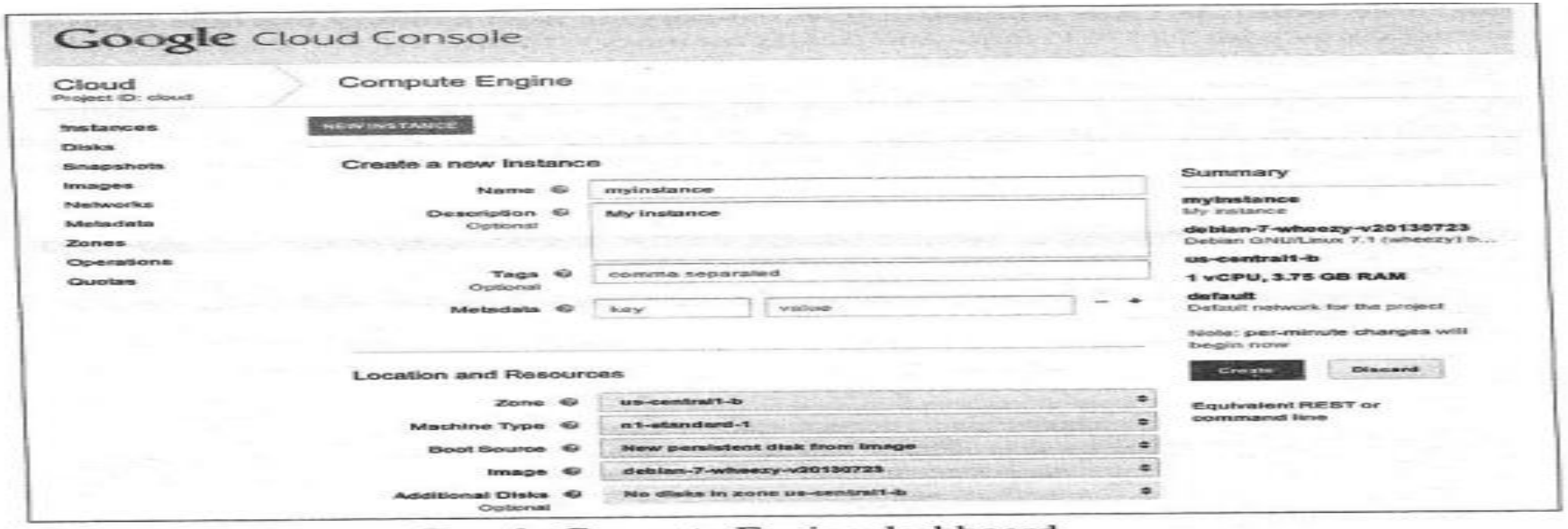

Amazon EC2 dashboard

# Cloud Services Examples:

## IaaS-Amazon EC2, Google Compute Engine, Azure VMs

## Google Compute Engine

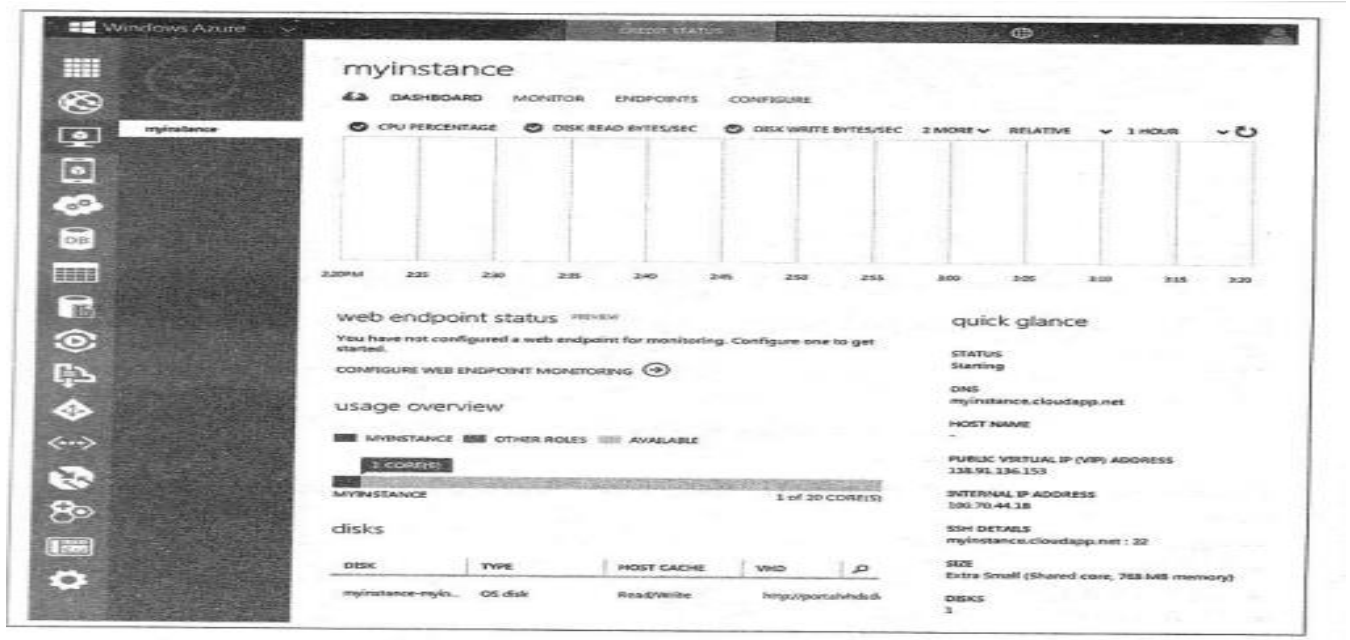
- **Google Compute Engine (GCE) is an IaaS offering from Google**.
- GCE provides **virtual machines of various computing capacities ranging from small instances** (Eg: Virtual core with 1.38 GCE unit and 1.7 GB memory) **to high memory machine types** (8 virtual cores with 22 GCE unit and 52 GB memory.
- The below figure shows screenshot of Google Compute Engine dashboard




Google Compute Engine dashboard

# Cloud Services Examples:
## IaaS-Amazon EC2, Google Compute Engine, Azure VMs
## Windows Azure

- **Windows Azure Virtual Machine is an IaaS offering from Microsoft**.
- Azure VMs provides **virtual machines of various computing capacities ranging from small instances** (1 virtual core with 1.75GB memory) to **memory intensive machine types** ( 8 virtual cores with 56GB memory).
- The below figure shows screenshot of Google Compute Engine dashboard.




Windows Azure Virtual Machines dashboard

# Cloud Services Examples:
## PaaS-Google App Engine

- **Google App Engine (GAE) is a Platform as a Service offering from Google**.

- GAE is a cloud based **web service for hosting web applications and storing data**.

- GAE allows users to **build scalable and reliable applications that run on the same systems that power Google's own applications**.

- GAE provides a **Software Development Kit (SDK) for developing web applications software that can be deployed on GAE.**

# Cloud Services Examples: PaaS-Google App Engine

- Developers can **develop and test their applications with GAE SDK on a local machine** and **then upload it to GAE** with a simple click of a button

- Applications hosted in GAE are **easy to build, maintain and scale**. Users **don't need to worry about launching additional computing instances when the applications load increases**.

- GAE provides **automatic scaling and load balancing capability**.

- GAE **supports applications written in several programming language**.

- With Java runtime environment developers can **build applications using Java programming language and standard Java technologies such as Java Servlets**. GAE also **provides runtime environment for Python programming languages**.

# Cloud Services Examples: PaaS-Google App Engine

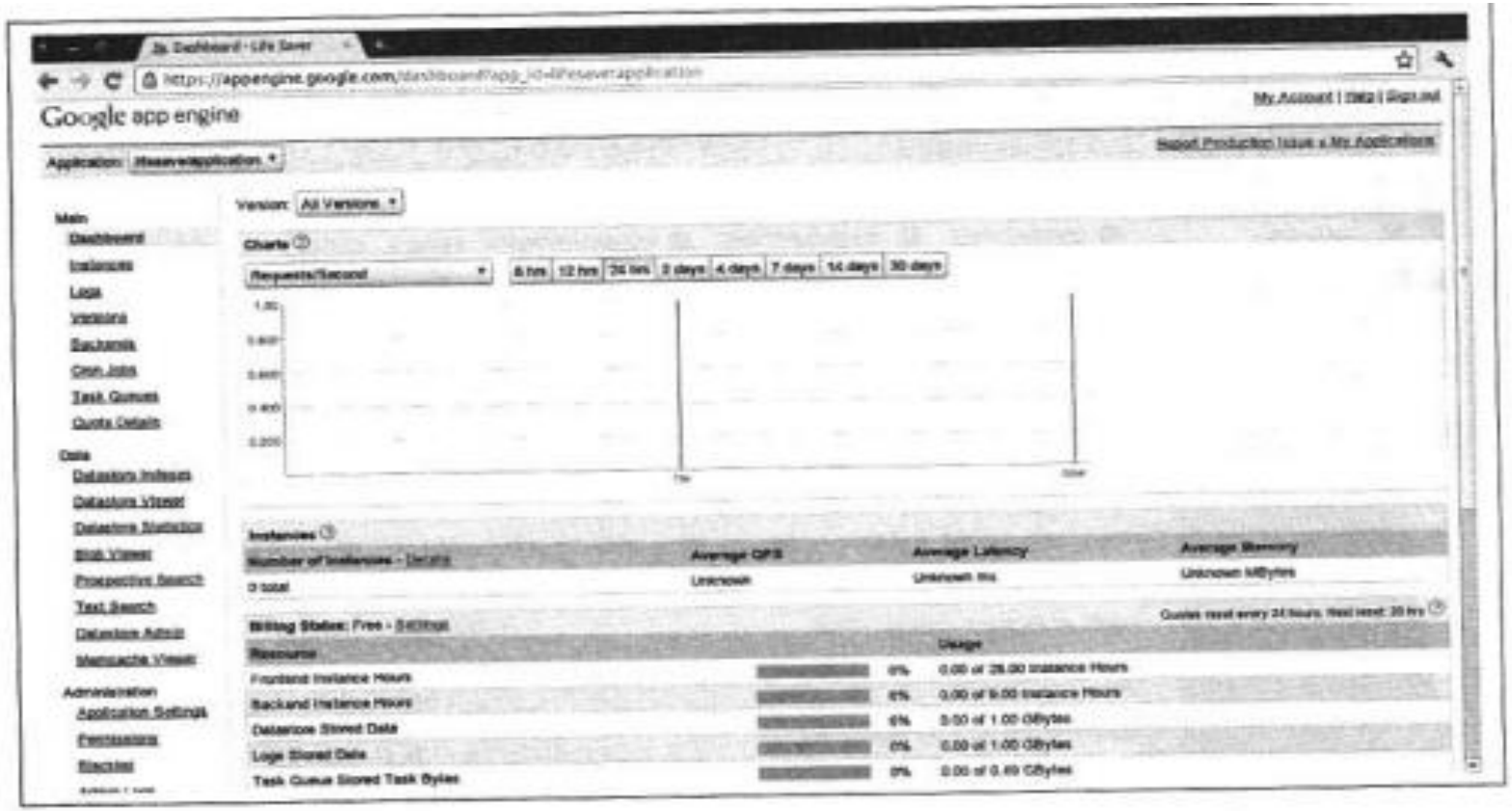
- **Applications hosted in GAE run in secure sandbox** with limited access to the underlying operating system and hardware.

- The **pricing model for GAE is based on the amount of computing resources used**.

- GAE provides **free computing resources for applications up to a certain limit**. **Beyond that limit, users are billed based on the amount of computing resources used such as amount of bandwidth consumed, number of resource instance hours, amount of data stored**.

# Cloud Services Examples: PaaS-Google App Engine

- The below figure shows the screenshot of GAE dashboard.




Google App Engine dashboard

# Cloud Services Examples: SaaS-Salesforce

**Salesforce Sales Cloud**

- Salesforce **Sales Cloud** is a cloud based **Customer Relationship Management (CRM) SaaS offering**.

- Users can **access CRM application from anywhere through internet enabled devices such as workstations, laptops, tablets and smartphones.**

- Sales Cloud allows **sales representatives to manage customer profiles**, **track opportunities**, **optimize campaigns from lead to close** and monitor the impact of campaigns. (A lead can be a company or an individual who has expressed interest in a company's product and/or service).

# Cloud Services Examples: SaaS-Salesforce

**Salesforce Service Cloud**

- **Salesforce Service Cloud is a cloud based Customer Service Management SaaS**.

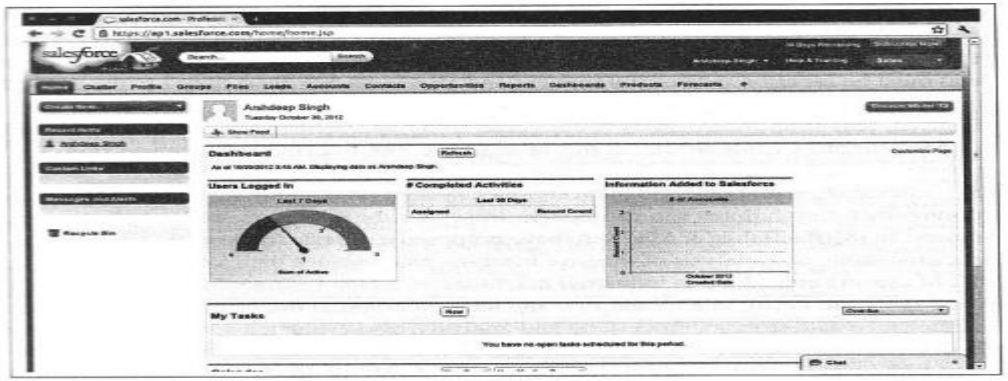
- Service cloud provides companies a call center like view and allows creating, tracking, routing and escalating cases.

- Service cloud includes a **social networking plug-in** that enables social customer service **where comments from social media channels can be used to answer customer questions**.

# Cloud Services Examples: SaaS-Salesforce

## Salesforce Marketing Cloud

- **Salesforce Marketing Cloud is cloud based social marketing SaaS**.
- Marketing cloud allows companies to **identify sales leads from social media**, discover advocates, identify most trending information on any topic.
- Marketing cloud **allows companies to pro-actively engage with customers**, manage **social advertisement campaigns** and track the performance of social campaigns.
- The below figure shows a screenshot of Salesforce dashboard




Salesforce dashboard

# Cloud Services Examples: SaaS-Salesforce

**Salesforce Marketing Cloud**

- **Some of the tools included in the Salesforce Sales, Service and Marketing Clouds include**
  - **Accounts and Contacts**
  - **Leads**
  - **Opportunities**
  - **Campaigns**
  - **Chatter**
  - **Analytics an Forecasts**

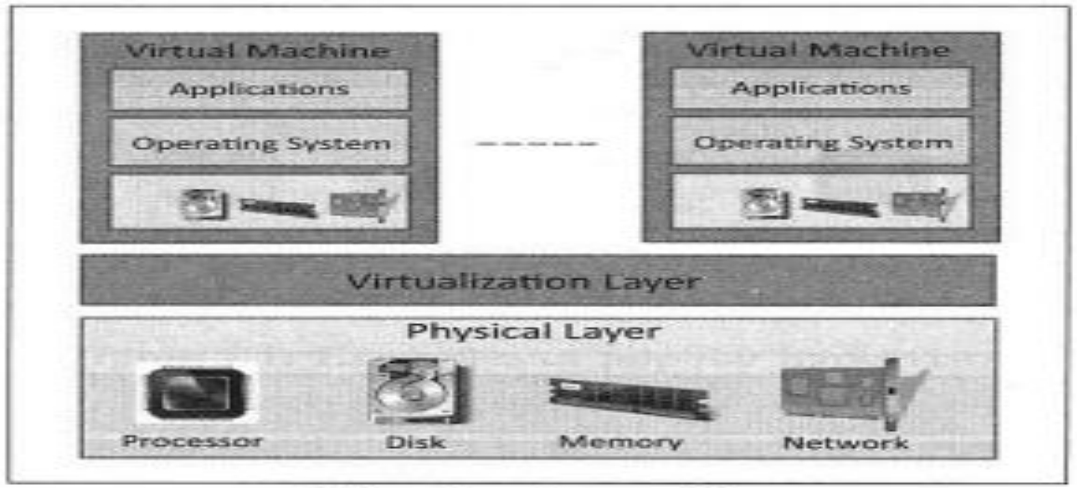
# Cloud Concepts and Technologies

- **Virtualization**
- **Load Balancing**
- **Scalability and Elasticity**
- **Deployment**
- **Replication**
- **Monitoring**
- **Software Defined Networking**
- **MapReduce**
- **Identity and Access Management**
- **Service Level Agreements**
- **Billing**

# Cloud Concepts and Technologies

## Virtualization

- Virtualization refers to **the partitioning the resources of a physical system** (such as computing, Storage, Network and Memory) **into multiple virtual resources**.
- In cloud computing, **resources are pooled to serve multiple users using Multi-Tenancy**.
- Multi-Tenant aspects of the cloud **allow multiple users to be served by the same physical hardware**.
- The below figure shows the architecture of a virtualization technology in cloud computing.
- The physical resources such as **computing, storage, memory and network resources are virtualized**.
- The **virtualization layer partitions the physical resources into multiple virtual machines**.

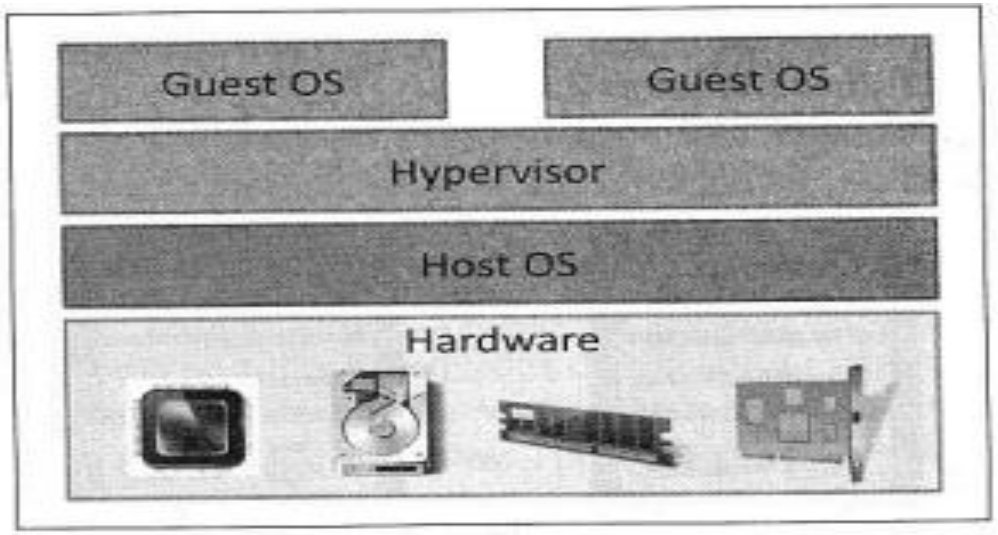



Virtualization architecture

# Cloud Concepts and Technologies

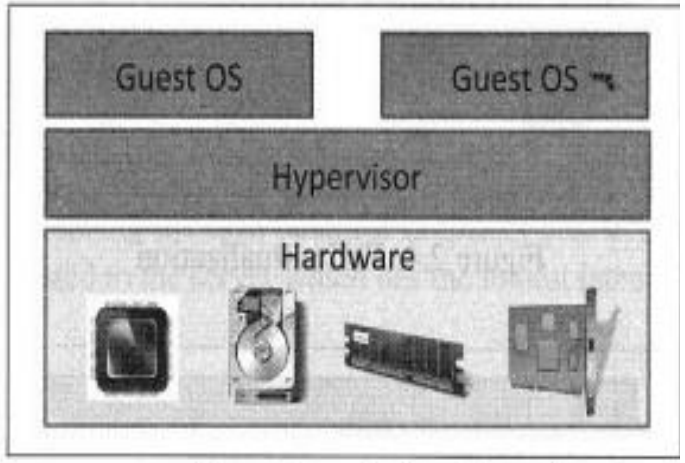
## Virtualization: Guest Operating System

- A **guest OS is an operating system that is installed in a virtual machine** in addition to the host OS.
- In virtualization, the **guest OS can be different from the host OS**.

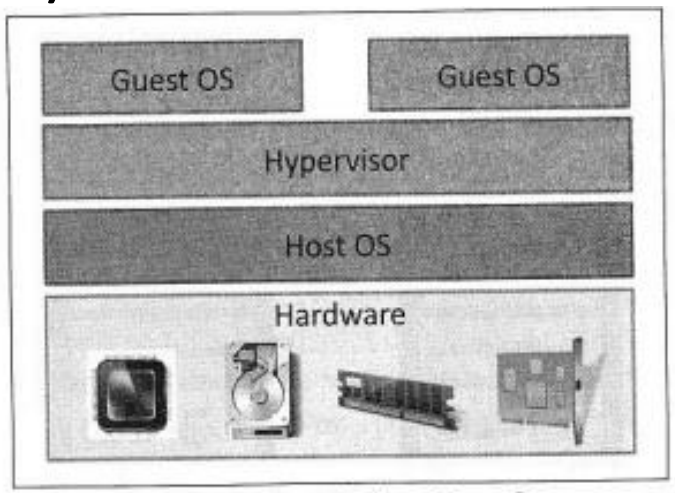


# Cloud Concepts and Technologies
# Virtualization: Hypervisor

- The **virtualization layer consists of a hypervisor or a Virtual Machine Monitor (VMM)**.
- There are two types of hypervisors
  - **Type-1 Hypervisors or Native Hypervisors**
  - **Type-2 Hypervisors or Hosted Hypervisors**



Hypervisor design: Type-1



Hypervisor design: Type-2

## Type-1 Hypervisors or Native Hypervisors

- **Type-1 Hypervisors or Native Hypervisors run directly on the host hardware** and **control the hardware** and **monitor the guest operating system**.

## Type 2 Hypervisors or Hosted Hypervisors

- **Type 2 Hypervisors or Hosted Hypervisors run on top of a conventional** (main or Host) **operating system** and **monitor the guest operation systems.**

# Cloud Concepts and Technologies

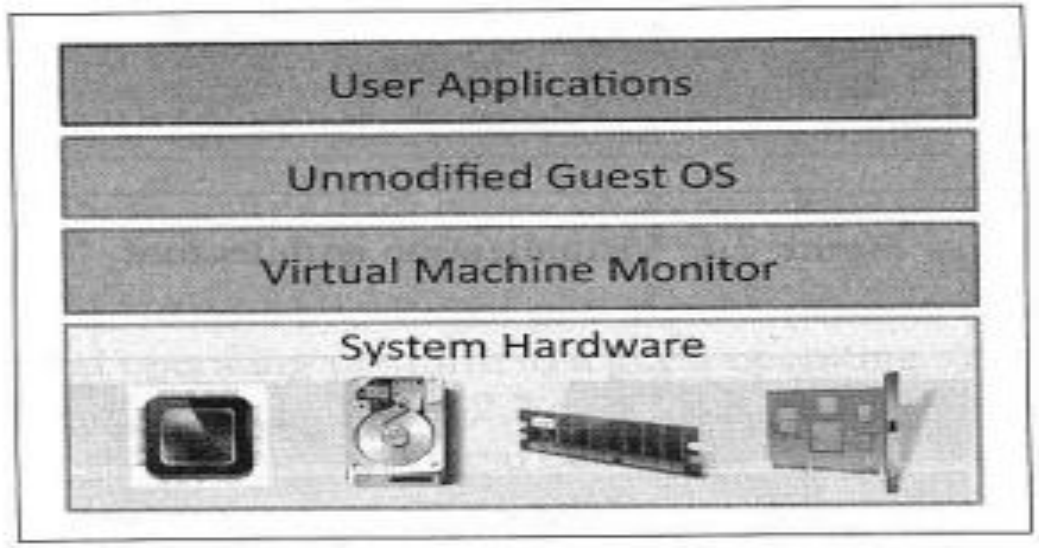
## Virtualization: Guest Operating System

- Various forms of virtualization approaches exist:
  - **Full Virtualization**
  - **Para-Virtualization**
  - **Hardware Virtualization**

# Cloud Concepts and Technologies

## Virtualization: Full Virtualization

- In Full Virtualization, **the guest OS requires no modification** and **is not aware that it is being virtualized**.
- **Full virtualization is enabled by direct execution of user requests** and **binary translation of OS requests**.
- **The below figure shows the Full virtualization approach**




Full virtualization

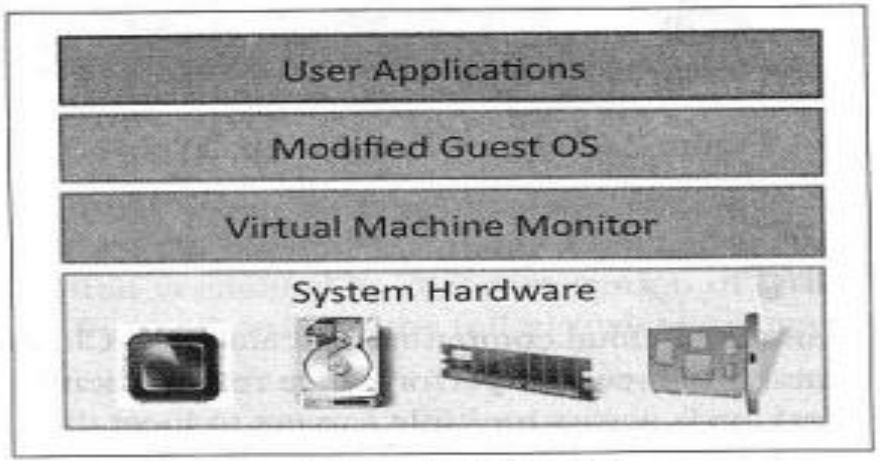
# Cloud Concepts and Technologies

## Virtualization: Para Virtualization

- In Para virtualization, **the guest OS is modified to enable communication with the hypervisor** to improve performance and efficiency.

- The **guest OS kernel is modified to replace non virtualizable instructions with hypercalls that communicate directly with the virtualization layer hypervisor.**

- The **below figure shows the para virtualization approach**




Para-virtualization

# Virtualization: Hardware Assisted Virtualization

- Hardware Assisted virtualization **is enabled by hardware features such as Intel's Virtualization technology** (VT-x) and AMD's **AMD-V**. In hardware virtualization, privileged and sensitive calls are set to automatically trap to the hypervisor.

- Thus, **there is no needed for either binary translation or Para virtualization**. Hardware-assisted full virtualization **eliminates the binary translation** and it **directly interrupts with hardware** using the virtualization technology which has been integrated on X86 processors since 2005 (Intel VT-x and AMD-V).

- **Guest OS's instructions might allow a virtual context execute privileged instructions directly on the processor, even though it is virtualized**.

# Cloud Concepts and Technologies

## Load Balancing Technique

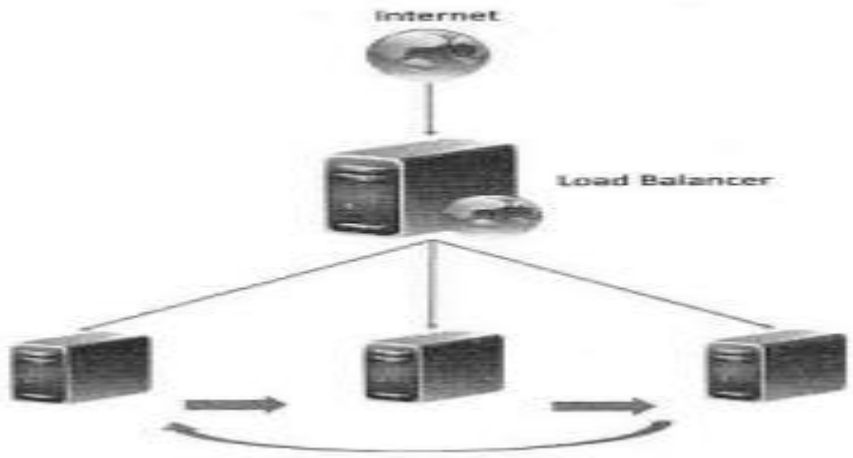
- One of the important features of cloud computing is **scalability**.
- Cloud resources can be **scaled up on demand to meet the performance requirements of applications**.
- **Load balancing distributes workload across multiple servers to meet the application workloads**.
- The **goal of load balancing techniques are to achieve maximum utilization of resources**, **minimize the response times, maximizing throughput**.
- Since multiple resources under a load balancer are used to serve the user requests, **in the event of failure of one or more of the resources**, the **load balancer can automatically reroute the user traffic to the healthy resources**.

# Cloud Concepts and Technologies

## Load Balancing Techniques

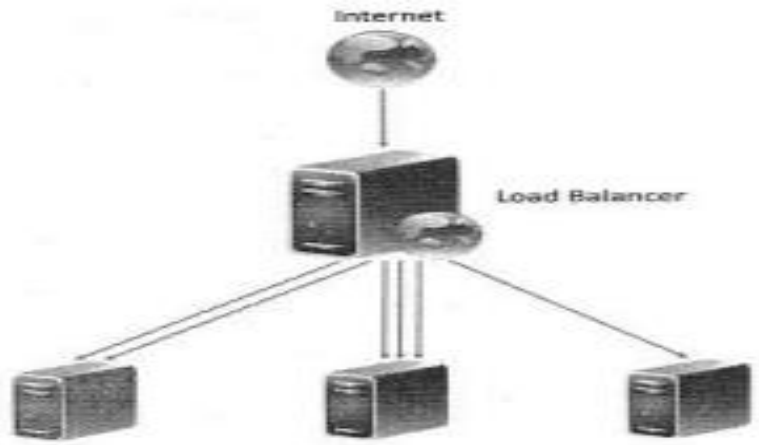
### Round Robin

- **In round robin load balancing, the servers are selected one by one in a circular fashion to server the incoming requests from the user.**




### Weighted Round Robin

- **In Weighted round robin load balancing, servers are assigned some weights. The incoming requests are proportionally routed to a server based on its weight.**

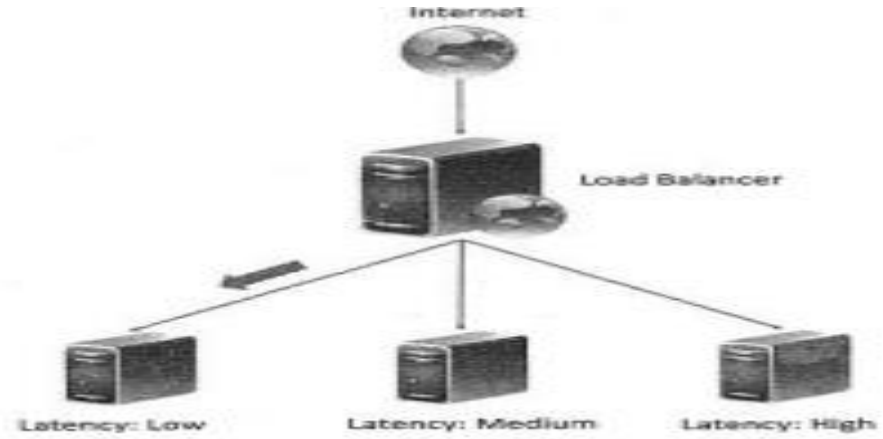


# Cloud Concepts and Technologies

## Load Balancing techniques

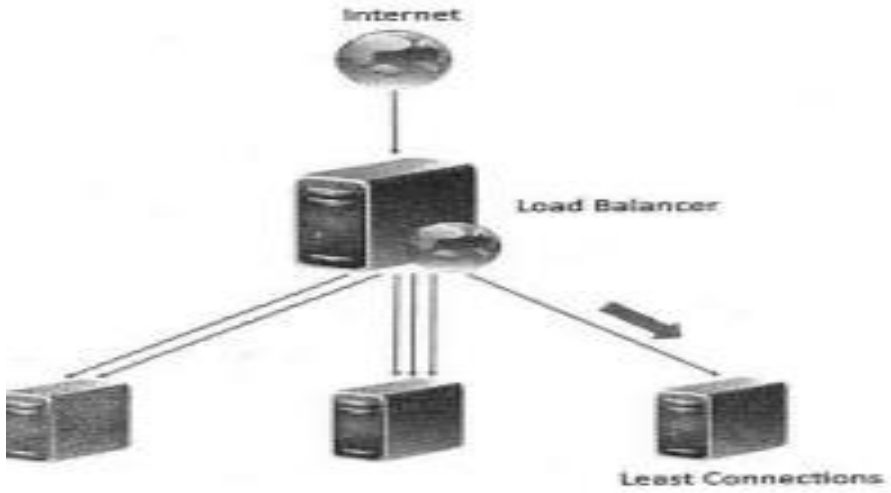
### Low Latency

- **In low latency load balancing, the load balancer monitors the latency of each server and request is routed to a server which has lowest latency.**




### Least Connections

- **In least connection load balancing, the incoming requests are routed to the server with least number of connections.**

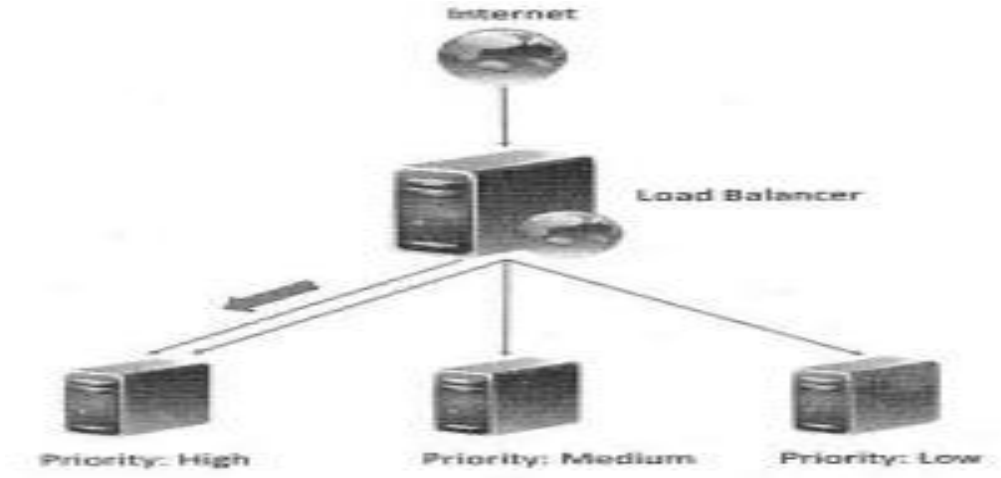


# Cloud Concepts and Technologies

## Load Balancing techniques

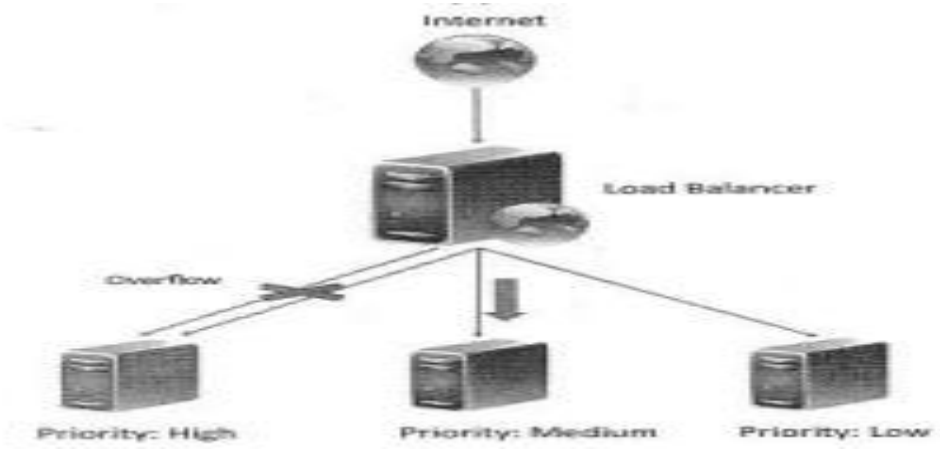
### Priority

- In priority load balancing, each server is assigned a priority. The incoming traffic is routed to the highest priority servers as long as the server is available. When the highest priority server fails, the incoming traffic is routed to a server with a lower priority.




### Overflow

- Overflow load balancing is similar to priority load balancing. When the incoming request to high priority servers overflow, the requests are routed to a lower priority server.



# Cloud Concepts and Technologies

## Load Balancing Techniques

- **A session is defined as a series of related browser requests that come from the same client during a certain time period.** Session tracking ties together a series of browser requests—think of these requests as pages—that may have some meaning as a whole, **such as a shopping cart application**.

- **A web session is a series of contiguous actions by a visitor on an individual website within a given time frame.** This could include your **search engine searches**, **filling out a form to receive content**, **scrolling on a website page**, **adding items to a shopping cart**, researching airfare, or which pages you viewed on a single website. **Any interaction that you have with a single website is recorded as a web session** to that website property.

- **To track sessions, a web session ID is stored in a visitor's browser**. This **session ID is passed along with any HTTP requests that the visitor makes while on the site** (e.g., clicking a link).

- ("Session" is the term used to refer to a visitor's time browsing a web site. It's meant to represent the **time between a visitor's first arrival at a page on the site and the time they stop using the site**.

- A **cookie is a small piece of data from a web site that is stored on a visitor's browser to help the website track the visitor's activity on the web site**.)

# Load Balancing techniques

- For the **session based applications, an important issue to handle during load balancing is the persistence of multiple requests from a particular user session.** (Because you may go from current page to next page and back to previous page)
- **Since load balancing can route successive requests from a user session to different servers, maintain the state or the information of the session is important**. Four persistence approaches are:
  - **Sticky Sessions**
  - **Session Database**
  - **Browser Cookies**
  - **URL Re-Writing**

# Cloud Concepts and Technologies

## Load Balancing Techniques

### Sticky Sessions

- In this approach, **all the requests belonging to a user session are routed to the same server**.
- **Theses sessions are called Sticky Sessions**.
- **The benefit of this approach is that it makes session management simple**.
- However, **a drawback of this approach is that if a server fails all the sessions belonging to that server are lost**, since there is **no automatic failover possible**.

### Session Database

- In this approach, **the session information is stored externally in a separate session database**, which is often replicated **to avoid a single point of failure**.
- **Though, this approach involves additional overhead of storing the session information**, however unlike the sticky session approach, **this approach allows automatic failover**.

# Cloud Concepts and Technologies

## Load Balancing Techniques

### Browser Cookies

- In this approach, **the session information is stored on the client side in the form of browser cookies**.
- **The benefit of this approach is that it makes the session management easy** and has the **least amount of overhead for the load balancer.**

### URL Re-Writing

- In this approach, a **URL re-write engine stores the session information by modifying the URL's on the client side**.
- Though this approach avoids overhead on the load balancer, **a drawback is that the amount of session information that can be stored is limited.** (Modifying each URL→ Shortening URL→ storing in engine→ degrades performance)
- **For applications that require larger amounts of session information, this approach does not work.**

(Changing a URL to the required format. URL rewriting allows URLs to be more easily remembered by the user. When the URL is entered into the Web server, the URL rewrite engine modifies the syntax behind the scenes to enable the appropriate Web page or database item to be retrieved. For example, to look up the definition for "path," a user friendly URL might look like computerlanguage.com/path. The rewrite engine could turn it into the following syntax: computerlanguage.com/results.php?definition=path. So, the URL rewrite function simply puts a layer on top of the original address and turns it into something easy to find and that makes sense. Thus, turning https://wiredelta.com/?page_id=16825 into wiredelta.com/url-rewrite, for example. From a user perspective, when a URL rewrite occurs the URL of the website remains the same in the browser and they are none the wiser. But behind the scenes, the browser rewrites the URL back into that complicated mess and sends a query to the servers.)

# Load Balancing Techniques

- **Load balancing can be implemented in software or hardware.**
- **Software based load balancers run on standard operating systems** and like other cloud resources.
- **Hardware load balancers implement load balancing algorithms in Application Specific Integrated Circuits** (ASICs).

# Cloud Concepts and Technologies

## Scalability and Elasticity

- **Muti-tier applications such as e-Commerce**, **Social networking**, **business-to-business** etc. **can experience rapid changes in their traffic.**

- **Each website has a different traffic pattern** which is determined by a number of factors that are **generally hard to predict beforehand.**

- **Capacity planning involves determining the right sizing of each tier of the deployment of an application in terms of number of resources and capacity of each resource**.

- **Capacity planning may be for computing, storage, memory or network resources**.

- Traditional approaches for capacity planning are based on **predicted demands for applications and account for worst case peak load as of applications.**

# Cloud Concepts and Technologies

## Scalability and Elasticity

- **When the workloads of applications increase, the traditional approaches have been either to scale up or scale out.**

- **Scaling up involves upgrading the hardware resources** (adding computing, memory, storage or network resources. **Scaling out involves addition of more resources of the same type**.

- Traditional scaling up and scaling out approaches are based on **demand forecasts at regular intervals of time**.

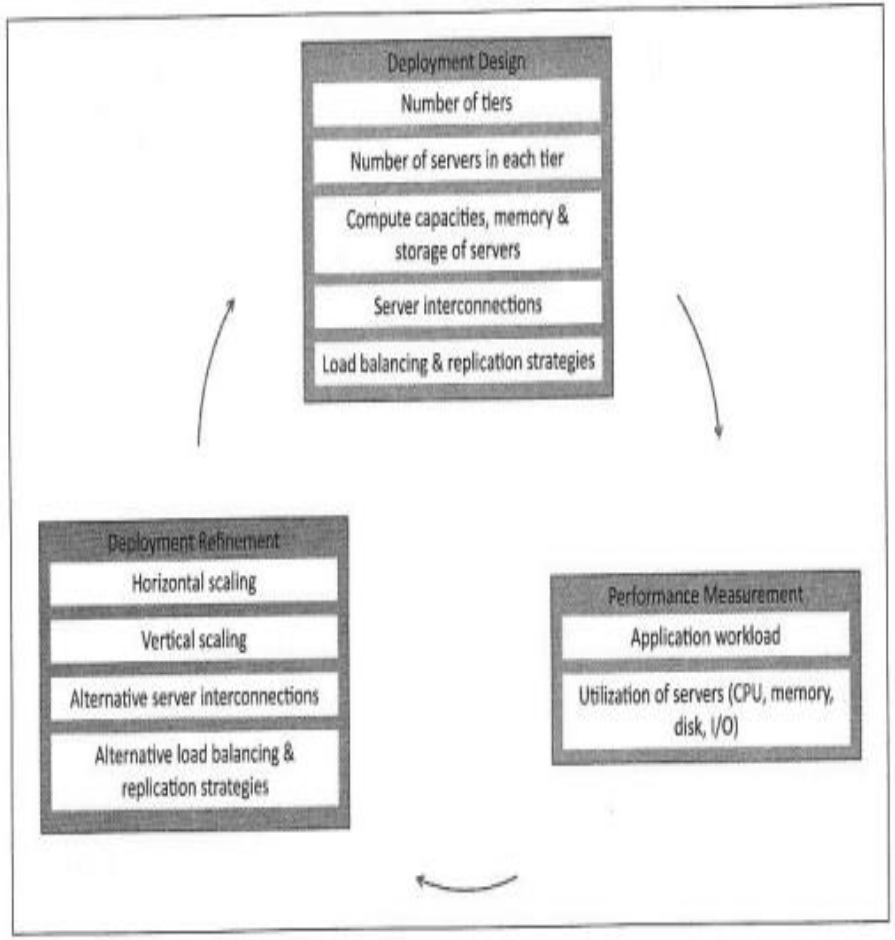
- When variations in workload are rapid, traditional approaches are **unable to keep track with the demand and either overprovisioning or under provisioning of resources**.

- **Over provisioning of resources of resources leads to higher capital expenditure** and **under provisioning of resources leads to traffic overloads, slow response time, low throughput.**

# Cloud Concepts and Technologies

## Deployment

- Figure shows the **cloud application deployment lifecycle**.
- Deployment prototyping can help in making architecture design choices.
- **By comparing performance of alternative deployment architectures**, deployment prototyping **can help in choosing the best and most effective deployment architecture that can meet the application performance requirements**.
- Deployment design is an iterative process that involves the following steps:
  - **Deployment Design**
  - **Performance Evaluation**
  - **Deployment Refinement**




Cloud application deployment lifecycle

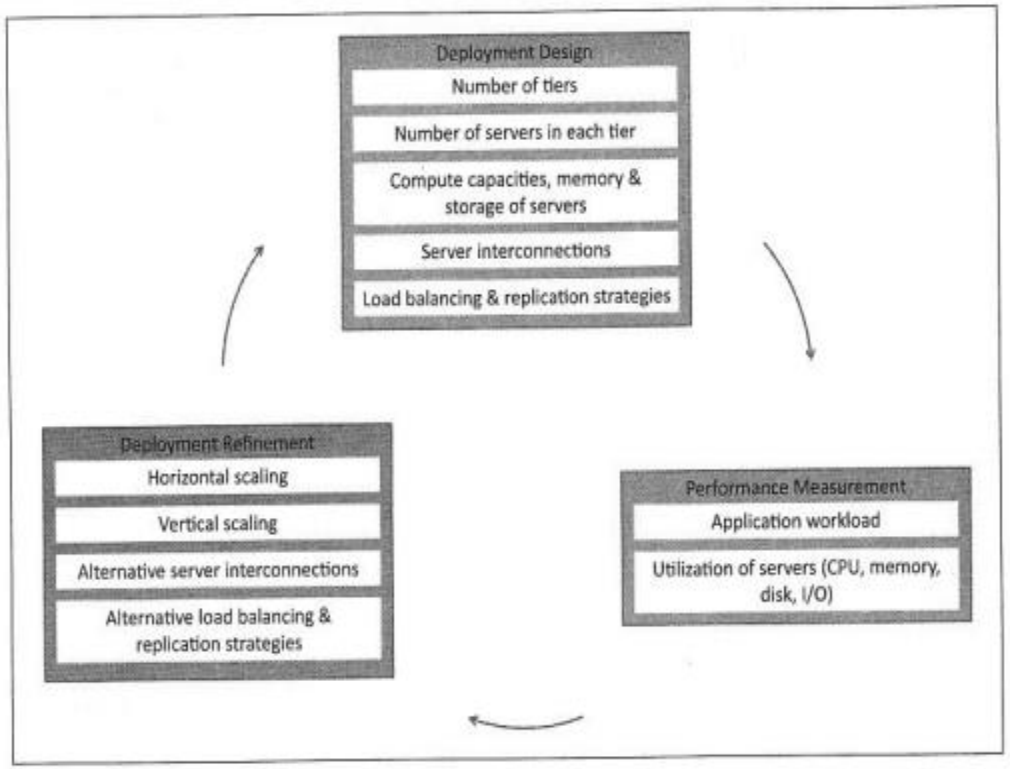
# Cloud Concepts and Technologies
# Deployment

## 1) Deployment Design

- In this step, the **application deployment is created with various tiers as specified in the deployment configuration**.

- The variables in this step include the **number of servers in each tier, computing, memory and storage capacities of servers, server interconnection, load balancing and replication strategies**.

- **Deployment is created by provisioning the cloud resources** as specified in the deployment configuration.




Cloud application deployment lifecycle

# Deployment

## 2) Performance Evaluation

- **Once the application is deployed in the cloud, the next step in the deployment lifecycle is to verify whether the application meets the performance requirements with the deployment.**

- This step involves **monitoring the workload on the application and measuring various workload parameters such as response time and throughput**. In addition, **the utilization of servers (CPU, memory, disk I/O etc.)** is also monitored.

## 3) Deployment Refinement

- **After evaluating the performance of the application, deployments are refined so that the application can meet the performance requirements.**

- Various alternatives can exits for **deployment refinement such as vertical scaling, horizontal scaling, alternative server interconnections, alternative load balancing and replication strategies**.

# Cloud Concepts and Technologies

## Deployment

- The below table lists some popular cloud deployment management tools.

| Cloud Deployment Management Tool | Features |
|---|---|
| RightScale | Design, deploy and manage cloud deployments across multiple public or private clouds. |
| Scalr | Provides tools to automate the management of servers, monitors servers, replaces servers that fail, provides auto scaling and backups. |
| Kaavo | Allows deploying applications easily across multiple clouds, managing distributed applications and automating high availability. |
| CloudStack | Allows simple and cost effective deployment management and configuration of cloud computing environments. |

Examples of popular cloud deployment management tools

# Cloud Concepts and Technologies

## Replication

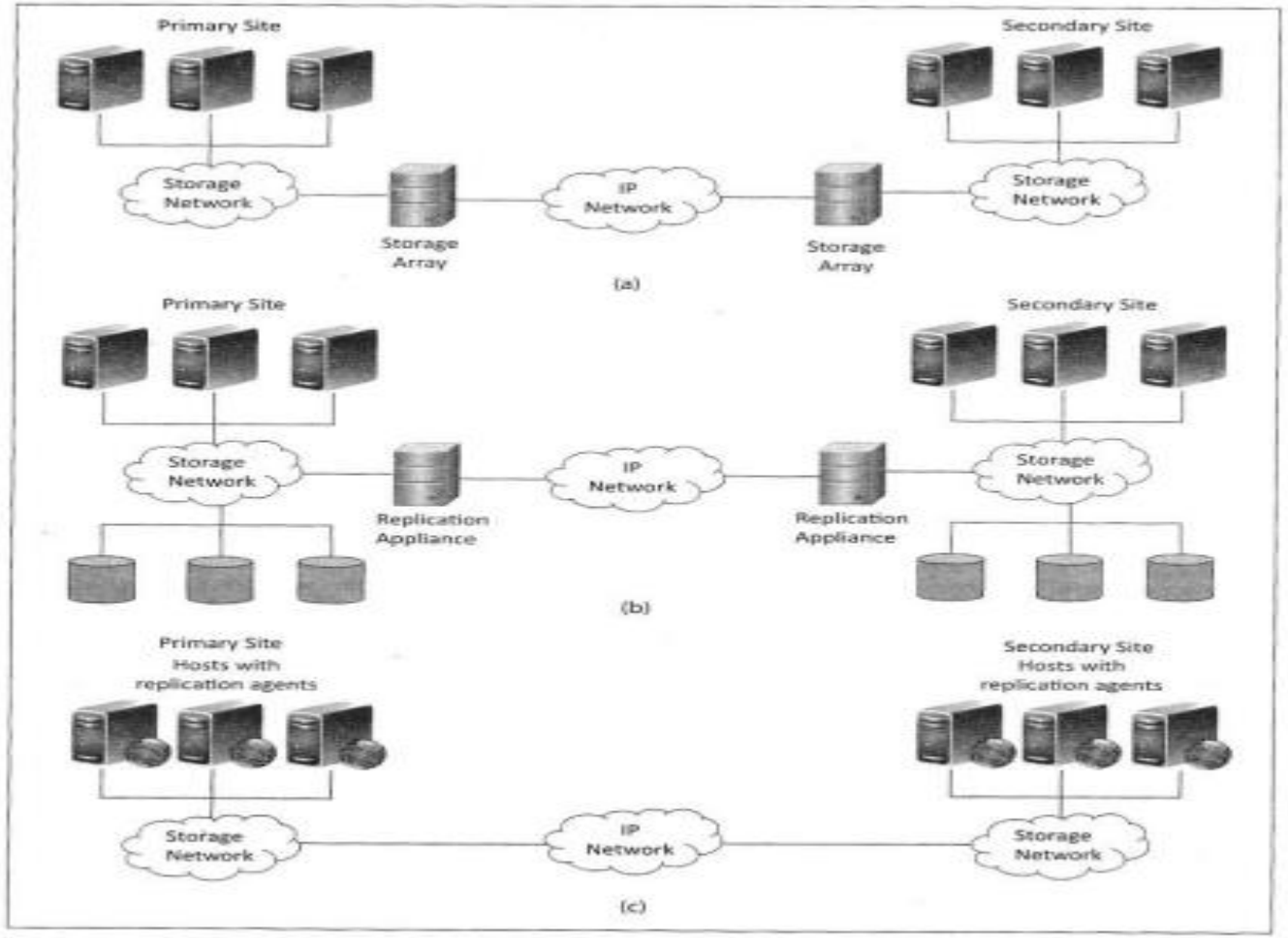
- Replication is used to **create and maintain multiple copies of the data in the cloud**.
- Replication of data is important for practical reasons **such as business continuity and disaster recovery**.
- **In the event of data loss at the primary location, organizations can continue to operate their applications from secondary data sources.**
- Traditional business time objective (RTO).
- continuity and disaster recovery approaches don't provide efficient, cost effective and automated recovery of data.
- **Cloud based data replication approaches provide replication of data in multiple locations, automated recovery, low recovery point objective (RPO)and low recovery time**.
- With **cloud based data replication**, **organizations can plan for disaster recovery without making any capital expenditures on purchasing, configuring and managing secondary site locations**.

# Cloud Concepts and Technologies

## Replication

- **There are three types of replication approaches as shown in below figure**

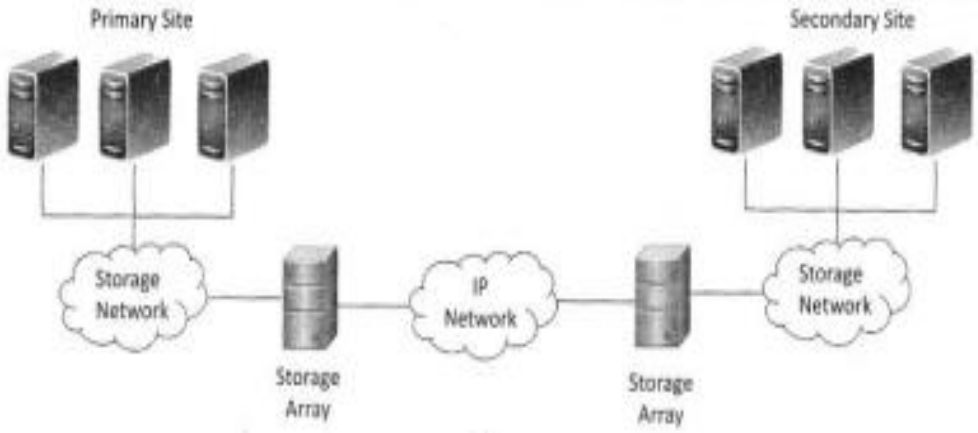
  - **Array Based Replication**
  - **Network Based Replication**
  - **Host Based Replication**




Replication approaches: (a) Array-based replication, (b) Network-based replication, (c) Host-based replication

# Cloud Concepts and Technologies

## Replication




### Array Based Replication

- Array Based Replication uses **compatible storage arrays to automatically copy data from a local storage array to a remote storage array**.

- **Arrays replicate data at the disk sub system level**(in terms of blocks), **therefore the type of hosts accessing the data and the type of data is not important**. **Thus array based replications can work in heterogeneous environments with different operating systems.**

- Array based replication uses **Network Attached Storage (NAS) or Storage Area Network (SAN)** to replicate.

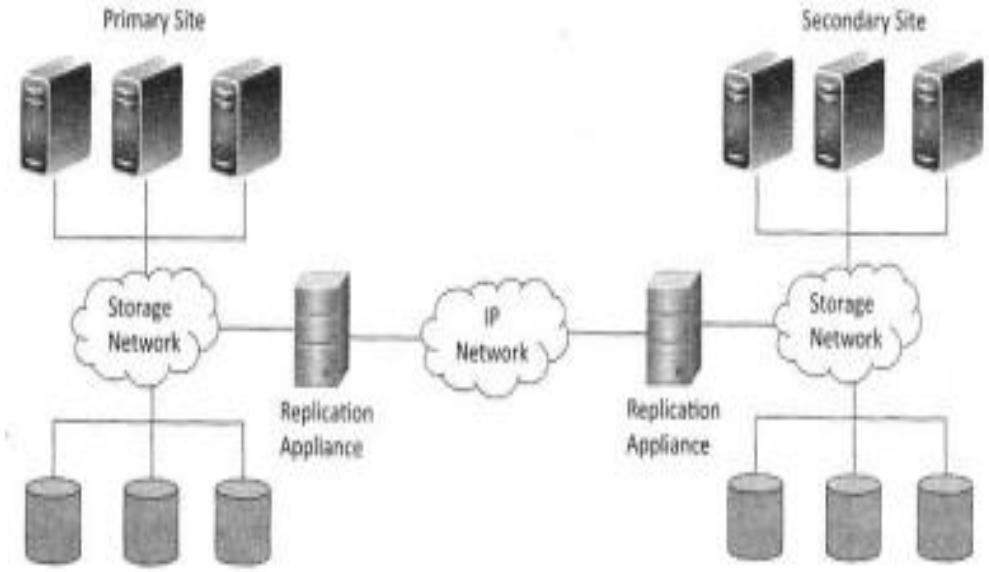
- **A drawback of this array based replication is that it requires similar arrays at local and remote locations.** (Storage methods, access methods, block size, block number, arrays etc must be of same). **Thus the costs for setting up array based replications are higher than the other approaches.**

- (Array-based replication is **an approach to data backup in which compatible storage arrays use built-in software to automatically copy data from one storage array to another**.)

# Cloud Concepts and Technologies

## Replication

### Network Based Replication




- **Network based replication uses an appliance or device that sits on the network** and **intercepts packets that are sent from hosts and storage arrays. The intercepted packets are replicated to a secondary locations.**

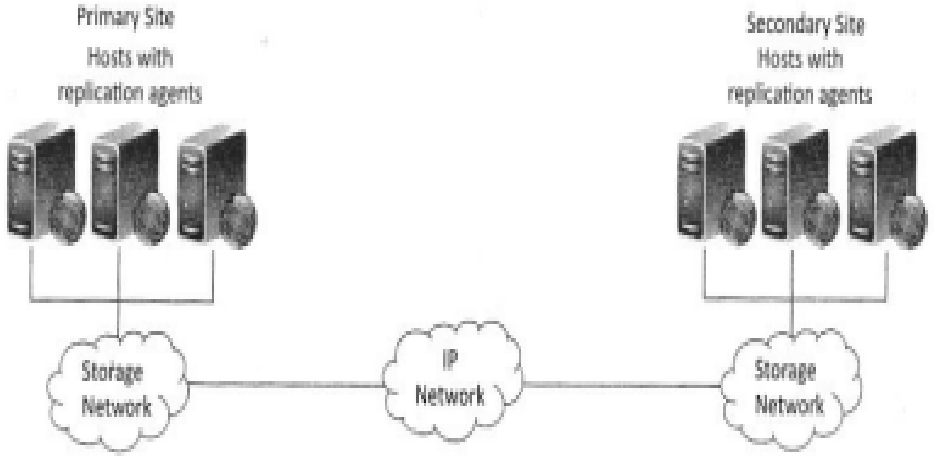
- **By offloading replication from server and storage systems, network-based replication can work across a large number of server platforms and storage systems, making it ideal for highly heterogeneous environments and requires a single point of management.**

- **This approach involves higher initial costs due to replication hardware and software.**

# Cloud Concepts and Technologies

# Replication

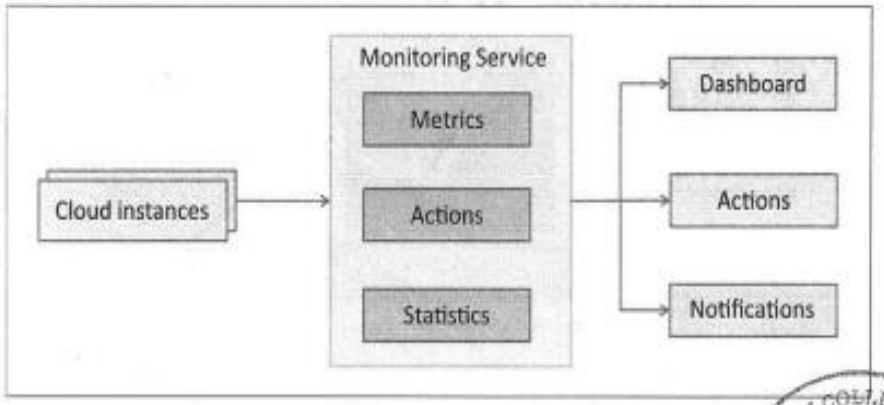
## Host Based Replication




- **Host based replication runs on standard servers and uses software to transfer data from a local to remote location.**
- **An agent is installed on the hosts that communicates with the agents on the other hosts**.
- **Host based replication can either be block based or file based**. Block based replication typically require dedicated volumes of the same size on both the local and remote servers.
- **With host based replication, entire virtual machines can be replicated in real time**.

# Cloud Concepts and Technologies

## Monitoring



Typical cloud monitoring service architecture

- **Cloud providers provides monitoring service that allows cloud users can monitor their cloud resource usage.**
- A monitoring service at the cloud **collects data on various system and application metrics from cloud computing instances.**
- **Users can define various actions based on the monitoring data**. For eg: **Auto scaling when the CPU usage of the monitored resources becomes high.**
- **Monitoring services also provides various statistics** based on the monitoring data collected as shown in below table.

| Type | Metrics |
|---|---|
| CPU | CPU-Usage, CPU-Idle |
| Disk | Disk-Usage, Bytes/sec (read/write), Operations/sec |
| Memory | Memory-Used, Memory-Free, Page-Cache |
| Interface | Packets/sec (incoming/outgoing), Octets/sec(incoming/outgoing) |

Typical monitoring metrics

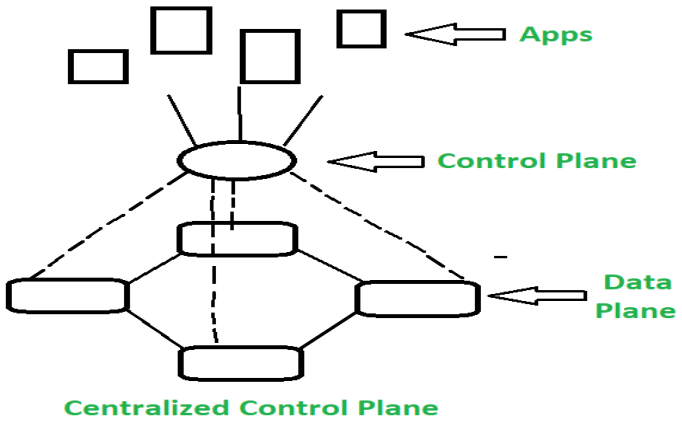
- **Monitoring of cloud resources is important because it allows the users to keep track of the health of applications and services deployed in the cloud.** For eg: an organization which has its website hosted in the cloud can monitor the performance of the website and also the website traffic with which users can make operational decisions such as scaling up or scaling down cloud resources.

# Cloud Concepts and Technologies

## Software Defined Networking

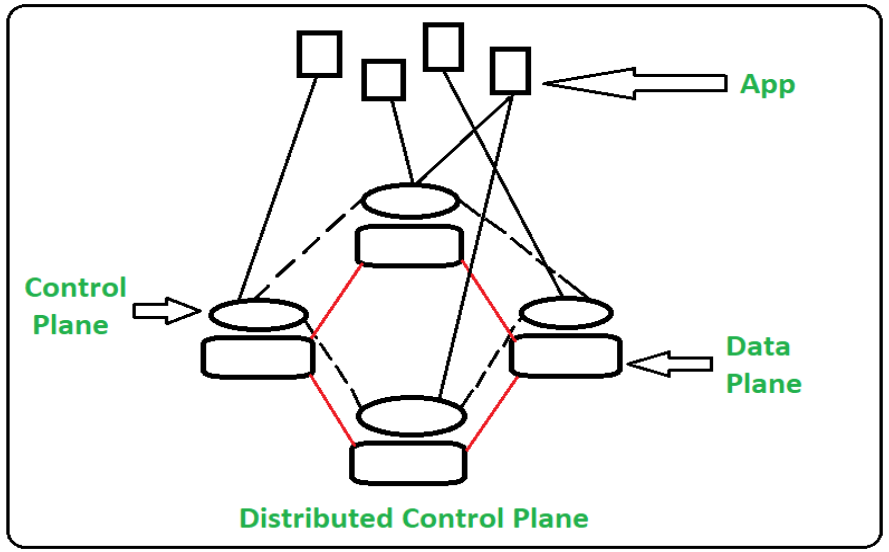
### Software Defined Networking

- **SDN decouples the network and control forwarding functions.**
- SDN stands for Software Defined Network which is networking architecture approach. It enables the control and management of network using software applications.
- Through Software Defined Network (SDN) networking **behavior of entire network and its devices are programmed in centrally controlled manner through software applications** using open APIs.




### Traditional Network

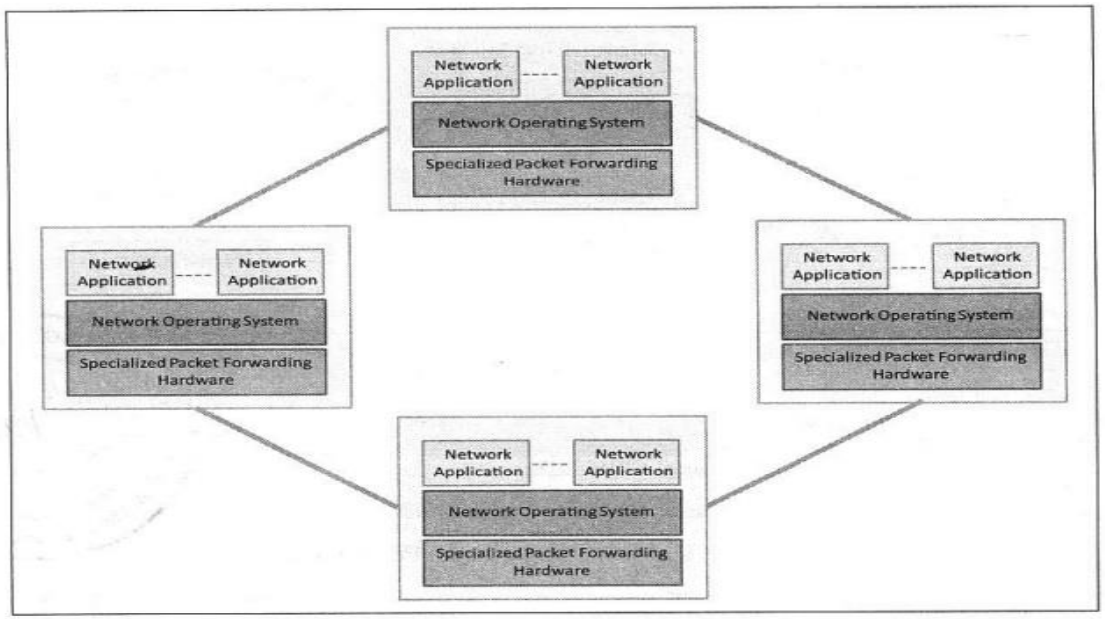
- **Traditional network refers to the old conventional way of networking which uses fixed and dedicated hardware devices such as routers and switches to control network traffic.**
- **Inability to scale and network security and performance are the major concern** now a days in the current growing business situation so that SDN is taking control to traditional network.
- **Traditional network is static** and based on hardware network appliances



# Cloud Concepts and Technologies

## Software Defined Networking

- **SDN decouples the network and control forwarding functions.** It separates the **control plane** (making traffic decision) from the **data plane** (packet forwarding).
- The below figure shows the conventional network architecture built with specialized hardware (Switches, routers etc).

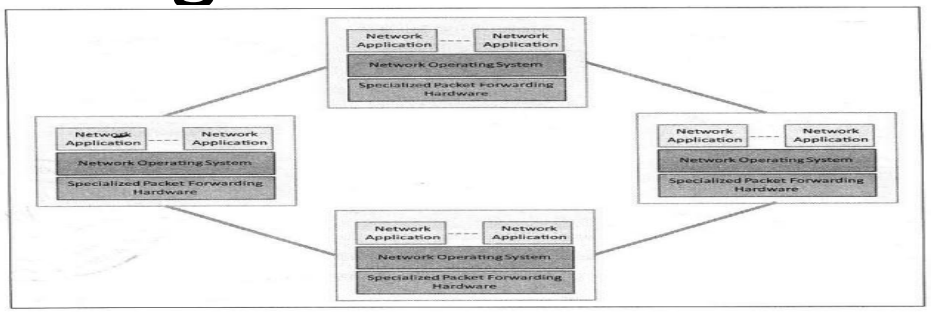



Conventional network architecture

- **In the conventional network architecture, the control plane and data plane are coupled**. **Control plane is the part of the network that carries the signaling and routing message traffic** while the **data plane is the part of the network that carries the payload data traffic.**

# Cloud Concepts and Technologies

## Software Defined Networking



Conventional network architecture

- **Conventional network architecture has following limitations**

### ➢ Complex Network Devices

**Conventional networks are getting increasingly complex with more and more protocols being implemented** to improve link speed and reliability. **The conventional networks are well suited for static traffic patterns**. **Due to the complexity of conventional network devices, making changes in the networks to meet the dynamic traffic patterns has increasingly difficult**.

### ➢ Management Overhead

**Network mangers find it increasingly difficult to manage multiple network devices**. **Upgradation of network requires configurations changes in multiple devices** (**switches, routers, firewalls** etc.)

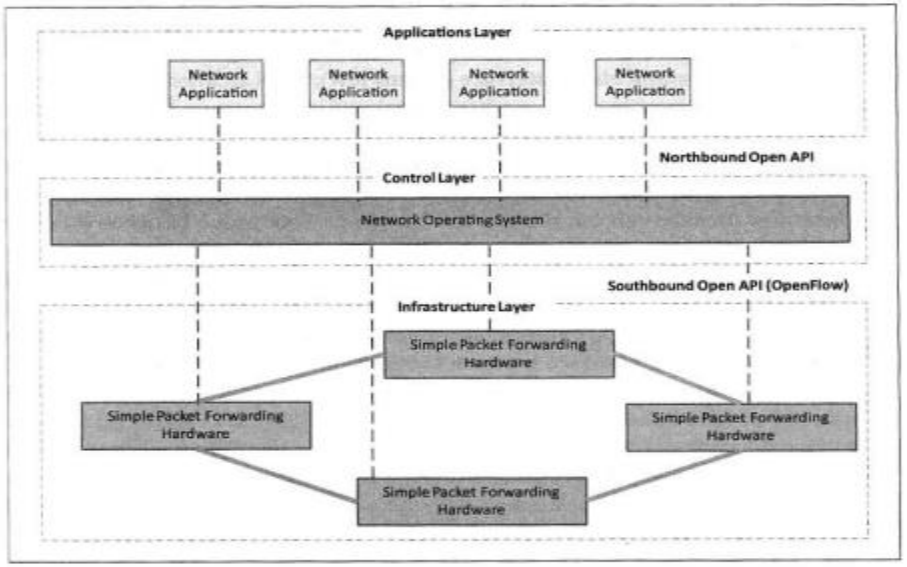
### ➢ Limited scalability

**Big data applications run distributed algorithms on a large number of virtual machines**. Such computing environments require highly scalable with minimum manual configurations which is difficult with conventional networks. (running each module of the applications at different places with multiple copies of data at different locations, need to scaling of the network)

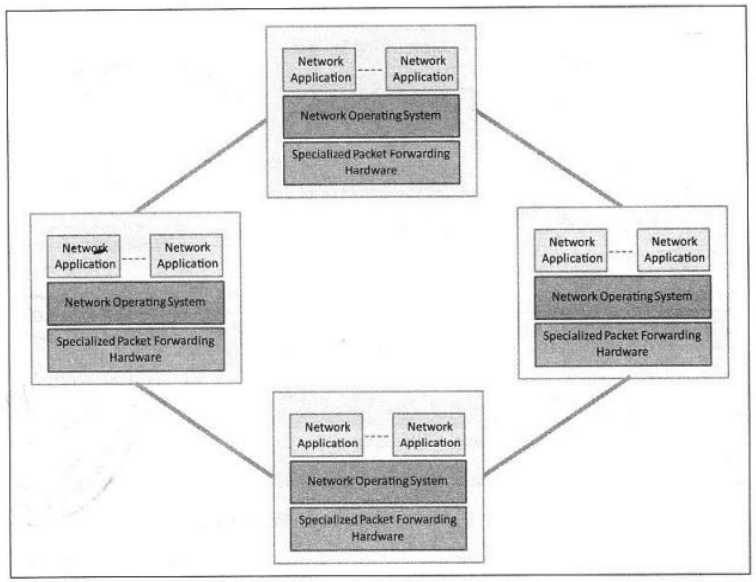
# Cloud Concepts and Technologies

## Software Defined Networking

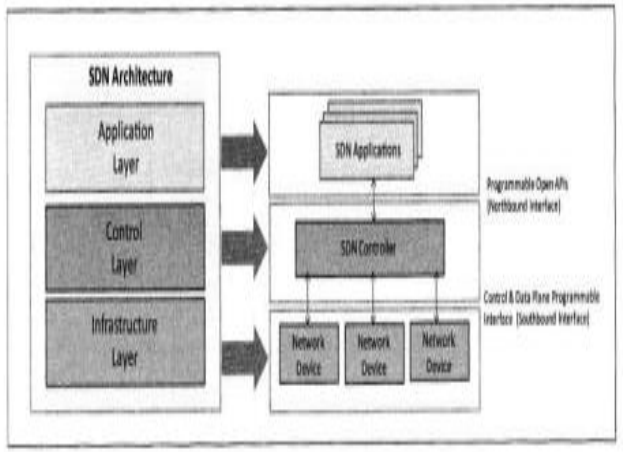
- **SDN attempts to create network architecture that are simpler, inexpensive, scalable, agile and easy to manage**.
- In SDN architecture, **the control plane and data planes are decoupled and network controller is centralized**.
- The **underlying infrastructure in SDN uses simple packet forwarding hardware as opposed to specialized hardware in conventional networks.**
- **Network devices become simple with SDN as they do not require implementation of large number of protocols**.
- **Network devices receive instructions from the SDN controller on how to forward the packets**.




SDN architecture




Conventional network architecture




SDN layers

# Cloud Concepts and Technologies

## Map Reduce

- **Map Reduce is a parallel data processing technique for processing and analyzing large scale data**. (data intensive problem). (The framework takes care of scheduling tasks, monitoring them and re-executing any failed tasks.)

- **The Map Reduce model includes two important phases: Map and Reduce functions.**

- **Map function takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (key/value pairs).** Secondly, **reduce function, which takes the output from a map as an input and combines those data tuples into a smaller set of tuples**.

# MAPREDUCE PROGRAMMING MODEL: Example taken from net for your reference

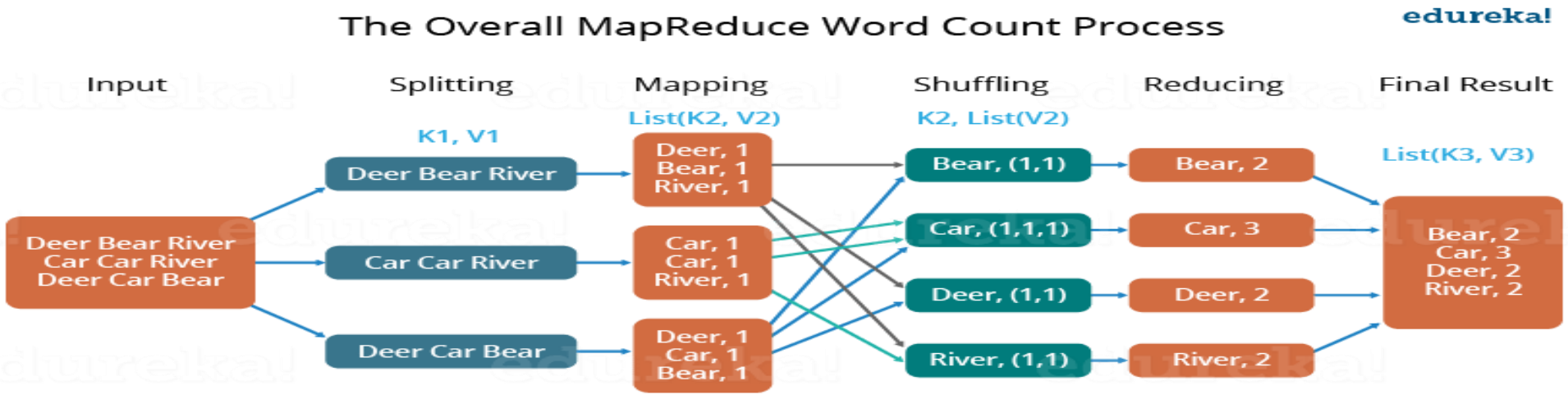
## A Word Count Example of MapReduce

Let us understand, how a MapReduce works by taking an example where there is a text file called **example.txt** whose contents are as follows:

**Dear, Bear, River, Car, Car, River, Deer, Car and Bear**

Now, suppose, we have to perform a word count on the **example.txt** using MapReduce to find the unique words and the number of occurrences of those unique words.

**Example taken from net for your reference**




- First, we divide the input in three splits as shown in the figure. This will distribute the work among all the map nodes. (Divides into file into 3 subfiles and each file given to each Map node.)

- Then, Map function within each Map node tokenize the words in the file and give a hardcoded value (1) to each of the tokens or words.

- Now, a list of key-value pair will be created where the key is nothing but the individual words and value is one. So, for the first line **(Dear Bear River)** we have 3 key-value pairs – **(Dear, 1); (Bear, 1); (River, 1)**. The mapping process remains the same on all the nodes.

- After mapper phase, a partition process takes place where sorting and shuffling happens so that all the tuples with the same key are sent to the corresponding reducer.

- So, after the sorting and shuffling phase, each reducer will have a unique key and a list of values corresponding to that key. For example, Bear, [1,1]; Car, [1,1,1].., etc. (Input to Reduce Node)

- Now, each Reducer counts the values which are present in that list of values. As shown in the figure, reducer gets a list of values which is [1,1] for the key Bear. Then, it counts the number of ones in the list and gives the final output as – Bear, 2.

- Finally, all the output key/value pairs are then collected and written in the output file.

# Cloud Concepts and Technologies

## Map Reduce ( for your reference)

- Among the cluster of nodes, one acts like a "**master**" and the rest are "**workers**."
- The **master is responsible for scheduling** (assigns the map and reduce tasks to the worker) and **monitoring** (monitors the task progress and the worker health). (Master instructs which nodes should act like a mapper and which nodes should act like a reducer. Also responsible for monitoring health condition of each worker node)
- When map tasks arise, the **master assigns** the task to an **idle worker**, taking into account the data locality.
- A **worker reads** the **content of the corresponding input split** and emits a **key/value pairs**. (On what operation performing and corresponding analysed value)
- The **intermediate key/value pairs** produced by the Map function are first **buffered in memory and then periodically written to a local disk**, **sorts** the intermediate keys so that **all occurrences of the same key are grouped together into R sets by the partitioning function**.
- The master passes the location of these stored pairs to the **reduce worker**, which **reads** the buffered data using remote procedure calls (**RPC**).
- For each key, the worker passes the corresponding intermediate value for its entire occurrence to the **Reduce function**. (worker node gives intermediate set to reduce function, which generates output file)
- Finally, the output is available in **R output files** (one per reduce task).

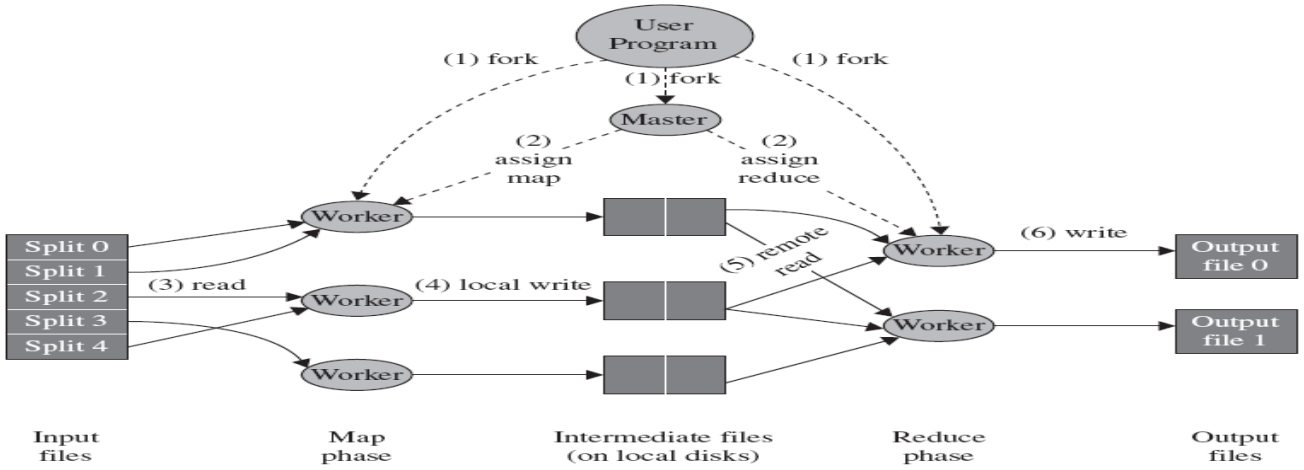



FIGURE 14.3. MapReduce execution overview [4].

# Cloud Concepts and Technologies

## Map Reduce

- **The below figure shows the workflow of MapReduce.**
- **In the Map phase, data is read from a distributed file system, partitioned among a set of computing nodes**.
- The **Map tasks process the input records independently of each other** and **produce intermediate results as Key-Value pairs**.
- **When all the Map tasks are completed**, the **Reduce phase begins** in which the **intermediate data with the same key is aggregated**.
- An **optional combine task can be used to perform data aggregation** on the intermediate data of the same key **for the output from the Mapper before transferring to the Reduce task**.

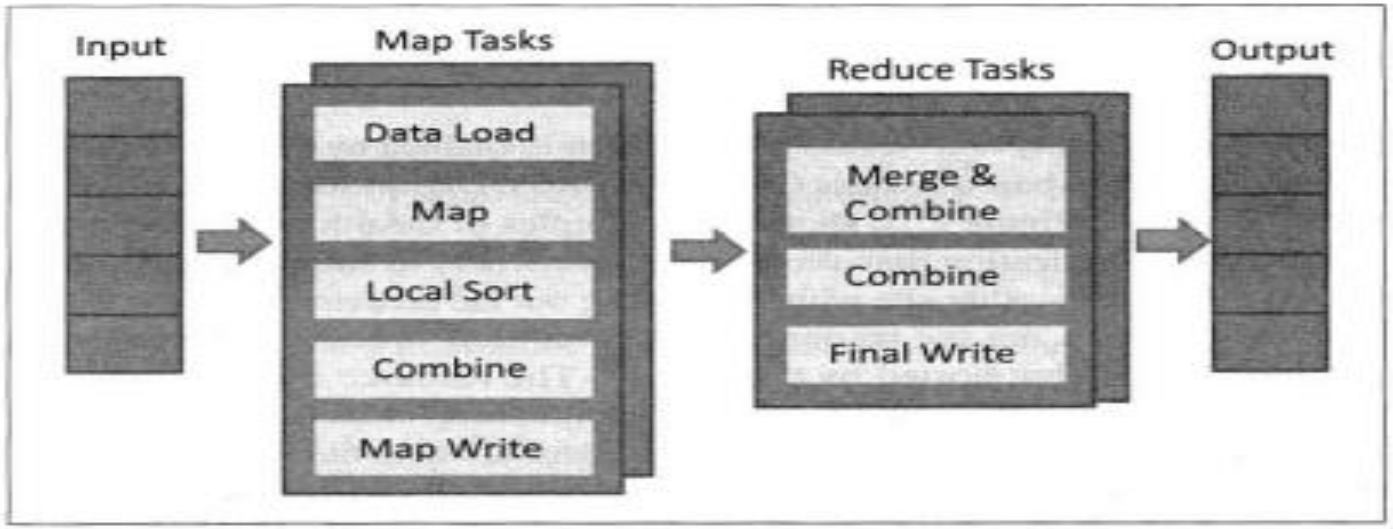



Figure : MapReduce workflow

# Cloud Concepts and Technologies

## Map Reduce

- **The below figure shows the flow of data for a MapReduce** job.

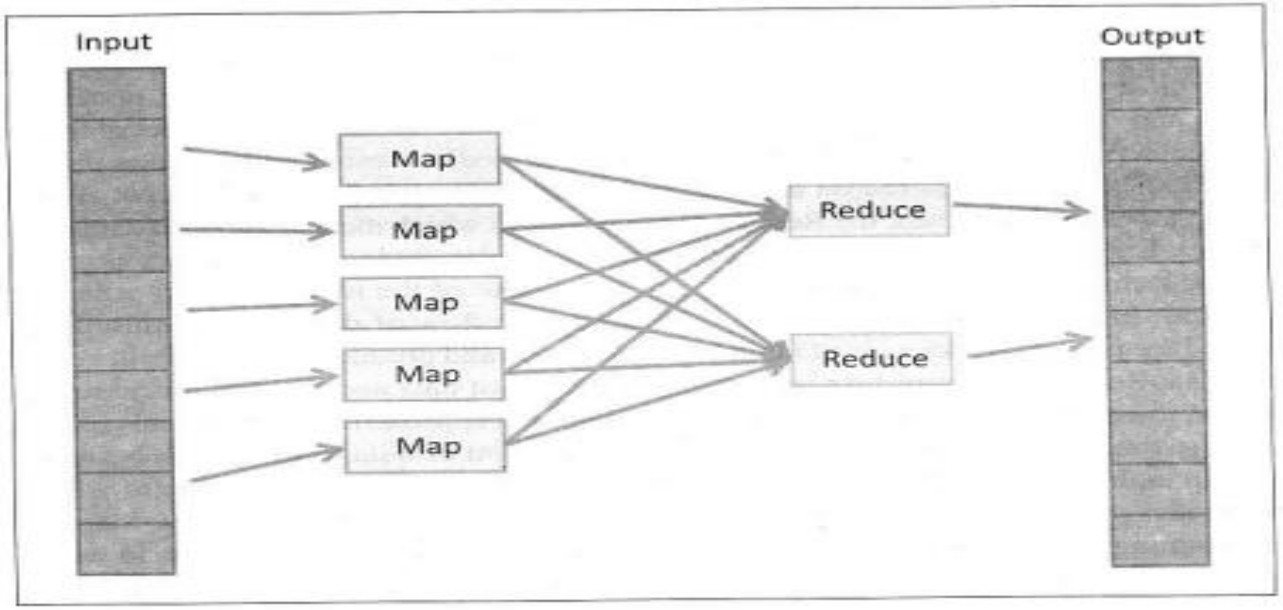



Figure Data flow in MapReduce

- **MapReduce model take advantage of locality of data** and the data processing takes place on the nodes where the data resides.
- **In traditional approach for data analysis, data is moved to the compute nodes** which results in significant of **data transmission between the nodes** whereas **MapReduce model moves the computation to where the data resides thus decreasing the transmission of data and improving efficiency**.

# Cloud Concepts and Technologies

## Identity and Access Management

- **Identity and Access Management (IDAM)** for cloud describes the **authentication and authorization of users to provide secure access to cloud resources.**

- Organization with multiple users can use IDAM services provided by the cloud service provider **for management of user identifiers and user permissions.**

- **IDAM services allow organizations to centrally manage users, access permission, security credentials and access keys.**

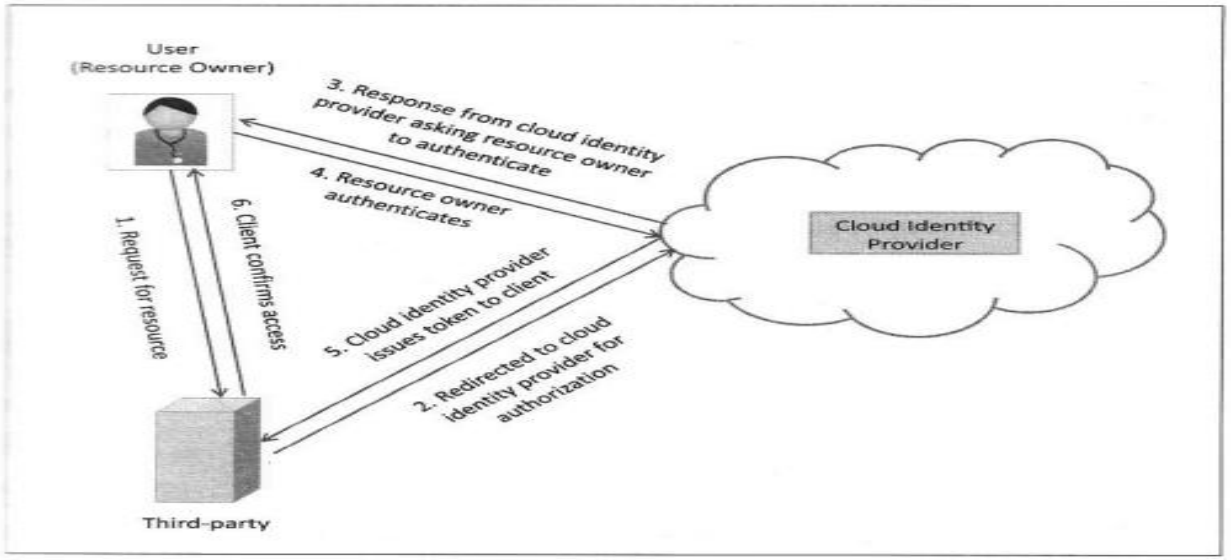
- IDAM services **allow creation of user groups where all the users in a group have the same access permissions.**

- Identity and Management is enabled by a number of technologies such as **OpenAuth, Role-based Access Control(RBAC), Digital Identities, Security Tokens, Identity Providers etc.**

# Cloud Concepts and Technologies

## Identity and access Management

- **The below figure shows the examples of OAuth identity and Access Management Service provided by the cloud service providers.**



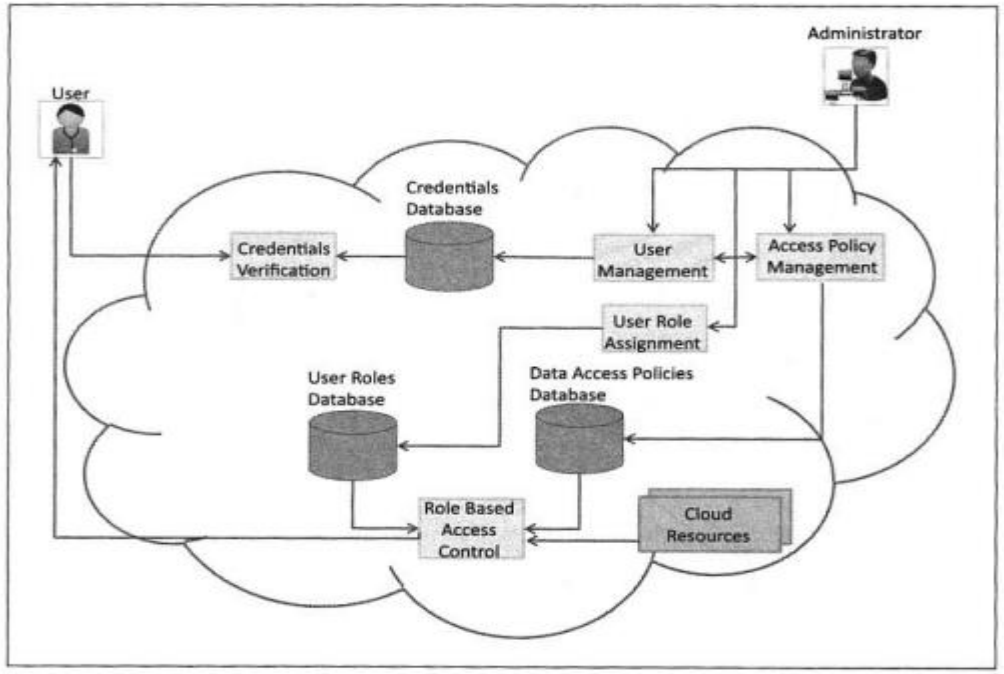

OAuth example

- **Oauth is an open standard for authorization that allows resource owners to share their private resources stored on one site with an other site without handling out the credentials.**
- In the Oauth model, **an application** (which is not the resource owner) **requests access to resources controlled by the resource owner** (but hosted by the server), **the resource owner grants permission to access the resources in the form of a token and matched shared-secret.**
- **Tokens can be issued with a restricted scope and limited lifetime and revoked independently.**

# Cloud Concepts and Technologies

## Identity and access Management

- **RBAC is an approach for restricting access to authorized users.**
- **The below figure shows an example of a typical RBAC framework.**
- **A user who wants to access cloud resources is required to send his/her data to the system administrator who assigns permissions and access control policies which are stored in the User Roles and Data Access Policies databases respectively.**




Role-based Access Control example

# Cloud Concepts and Technologies

## Service Level Agreement

- **A Service Level Agreement (SLA)** for cloud **specifies the level of service that is formally defined as a part of the service contract with the cloud service provider.**

- **SLAs provide a level of service for each service** which is **specified in the form of minimum level of service guaranteed and a target level.**

- **SLAs contain a number of performance metrics and the corresponding service objectives.**

- **The below table lis**

| Criteria | Details |
|---|---|
| Availability | Percentage of time the service is guaranteed to be available |
| Performance | Response time, Throughput |
| Disaster Recovery | Mean time to recover |
| Problem resolution | Process to identify problems, support options, resolution expectations |
| Security and privacy of data | Mechanisms for security of data in storage and transmission |

Table : List of criteria for cloud SLAs

# Cloud Concepts and Technologies

## Billing

- **Cloud Service providers offer a number of billing models described as follows:**
  - **Elastic Pricing**
  - **Fixed Pricing**
  - **Spot Pricing**

## Elastic Pricing

- **In Elastic pricing** or **Pay-as-you-use pricing model,** the **customers are charged based on the usage of cloud resources.**
- **Cloud computing provides the benefit of provision resources on demand.**
- **On-demand provisioning and elastic pricing models bring cost savings for the customers.**
- Elastic pricing model is **suited for customers who consume cloud resources for the short durations and who cannot predict the usage beforehand.**

# Cloud Concepts and Technologies

## Billing

### Fixed Pricing

- **In fixed pricing models, customers are charged a fixed amount per month for the cloud resources.**
- For eg: Fixed amount can be **charged per month for the running a virtual machine instance**, **irrespective of the actual usage.**
- Fixed pricing model is **suited for customers who want to use cloud resources for longer durations and want more control over the cloud expenses.**

### Spot Pricing

- **Spot pricing models offer variables pricing for the cloud resources which is driven by market demand.**
- **When the demand for the cloud resources is high, the prices increase** and **when the demand is lower, the prices decrease.**
- **The below table lists the billable resources for cloud including virtual machines, network., storage, data services, security services, support, application services, deployment and management services.**

| Resource | Details |
|---|---|
| Virtual machines | CPU, memory, storage, disk I/O, network I/O |
| Network | Network I/O, load balancers, DNS, firewall, VPN |
| Storage | Cloud storage, storage volumes, storage gateway |
| Data services | Data import/export services, data encryption, data compression, data backup, data redundancy, content delivery |
| Security services | Identity and access management, isolation, compliance |
| Support | Level of support, SLA, fault tolerance |
| Application services | Queuing service, notification service, workflow service, payment service |
| Deployment and management services | Monitoring service, deployment service |

Table : List of billable resources for cloud

# Cloud Based Services and Applications

- **Some examples on cloud based services and applications are:**

## Cloud Computing for Healthcare

- The figure shows the application of cloud computing environments to the healthcare ecosystem.
- Hospitals and their affiliated providers can securely access patient data stored in the cloud and share the data with the other hospitals and physicians.
- Patients can access their own health information from all of their care providers and store it in a personal health record (PHR)
- The PHR can be a vehicle for e-prescribing, a technique known to
- reduce medication dispensing errors and to facilitate medication reconciliation.
- History and information stored in the cloud can be streamline the admissions, care and discharege process by eliminating redundant data collection and entry.
- Health payers can increase the effectiveness and lower the cost of their care management programs by providing value added services and giving access to health information to members.

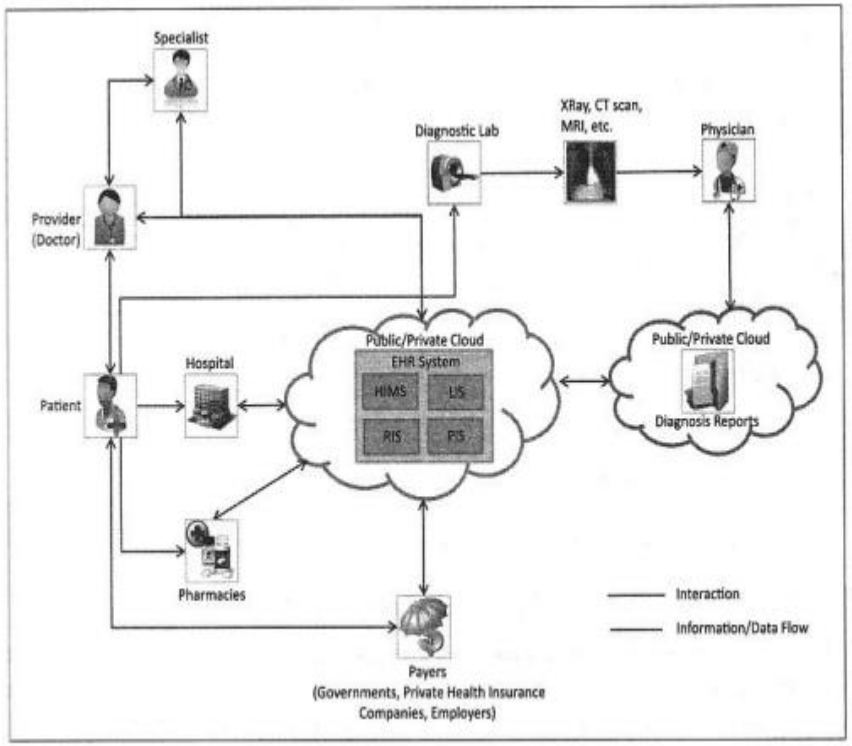



Figure : Cloud computing for healthcare

# Cloud Based Services and Applications

## Cloud Computing for Energy Systems

- Energy systems such as Smart Grids, Power Plants, Wind Turbine etc. have thousands of sensors that gather real time maintenance data continuously for condition monitoring and failure prediction purposes.
- The energy systems have a large number of critical components that must function correctly so that the systems can perform their operations correctly.
- For eg: a wind turbine has number of critical components like bearings, turning gears etc. that must be monitored carefully as wear and rear in such critical components or sudden change in operating conditions of the machines can result in failures. In systems such as power grids, real time information is collected using specialized electrical sensors called Phasor Measurement Units (PMU) at the sub-stations. The information received from PMUs must be monitored in real time for estimating the state of the system and for predicting failures.
- There is a generic framework "CloudView" for storage, processing and analysis of massive machine maintenance data collected from a large number of sensors embedded in industrial machines in a cloud computing environment.
- The below figure shows a generic use case of cloud for energy systems.

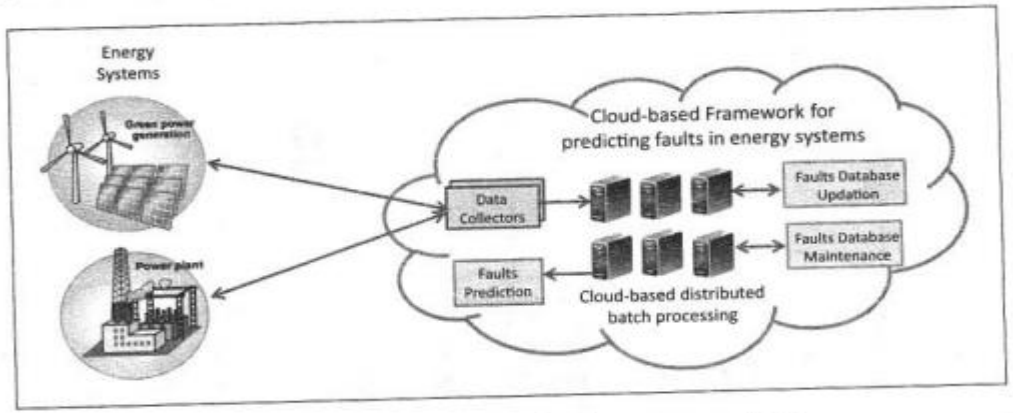



Figure : Cloud computing for energy systems

# Cloud Based Services and Applications

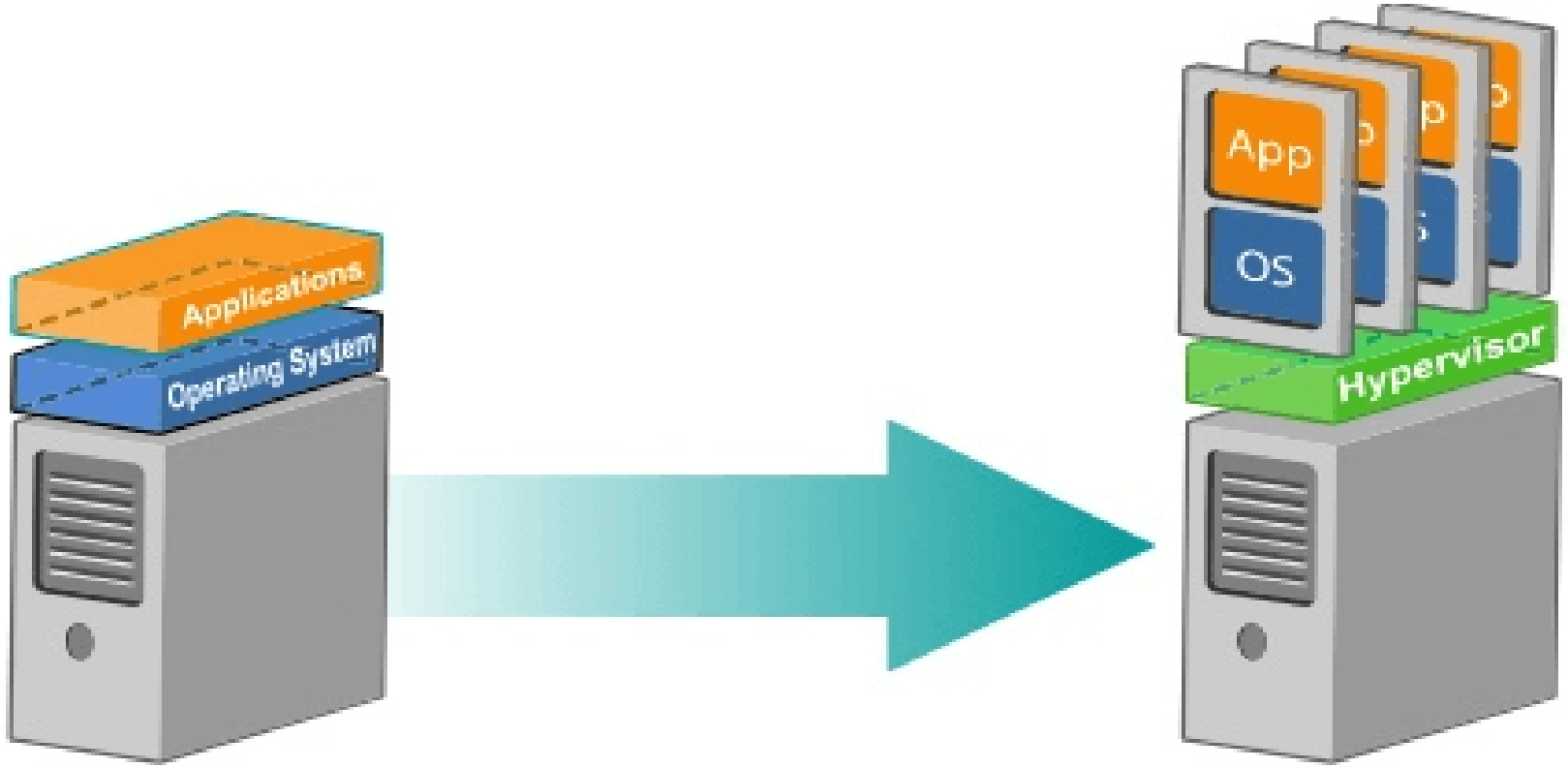
- **Cloud Computing for Transportation System**

# Unit-2
# VIRTUALIZATION FOR CLOUD

# Virtualization for Cloud

- **Virtualization is a technology** that makes it possible to **run multiple applications and various operating systems on the same server at the same time**.

- It increases **hardware utilization, saves energy and costs**.



# NEED FOR VIRTUALIZATION

- **Virtualization provides various benefits** including **saving time** and **energy**, **decreasing costs** and **minimizing overall risk.** (Saves Time: instead of maintaining 3 or 4 servers, maintain single server, configuring single server, saves power energy, costs, managing single server→ reduces overall risks)

- **Provides ability to manage resources effectively**. (Admin can concentrate on managing few number of servers)

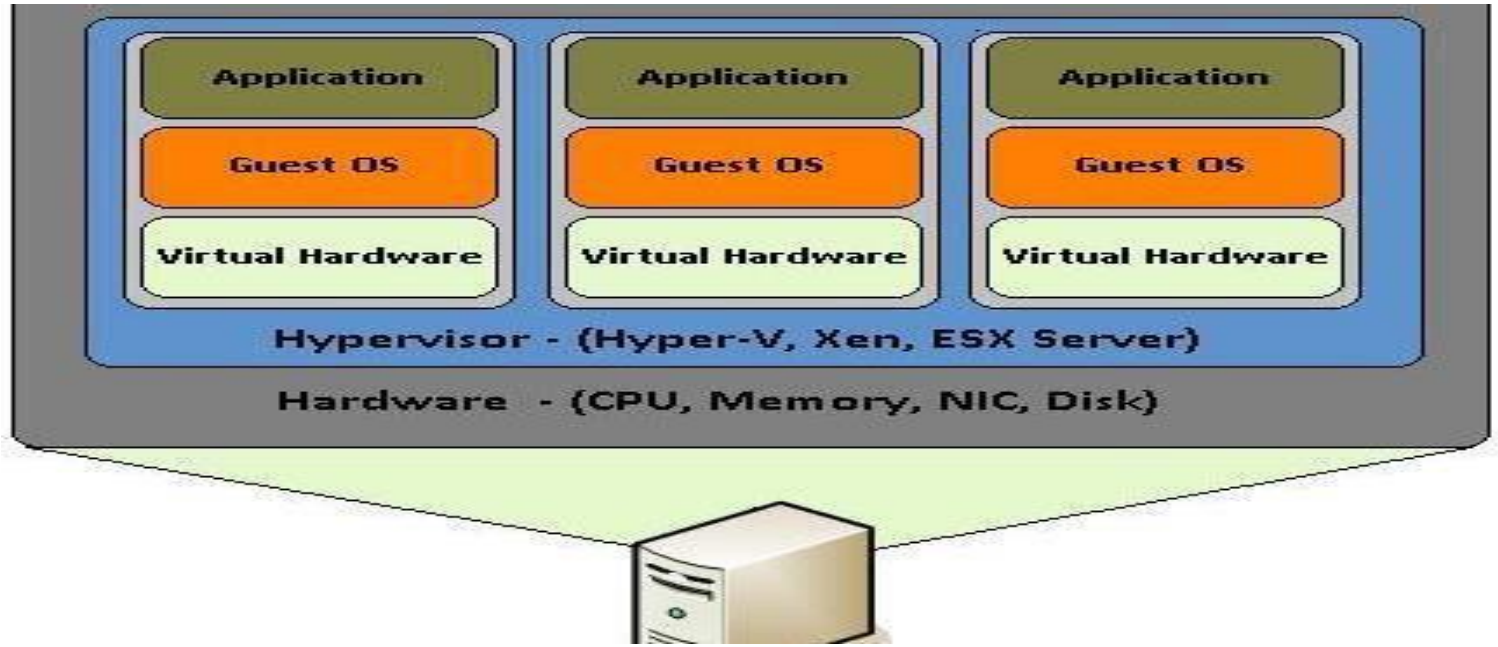
- **Increases productivity, as it provides secure remote access**. (More number of users can access their own VM from a single server in more secured way. Complete VM is given, can access within organization, while roaming, or at home in secured way)

- **Provides for data loss prevention**. (No data leakage between the users, Complete control on individual VM and complete VM is given in a secured way)

# What makes virtualization possible?

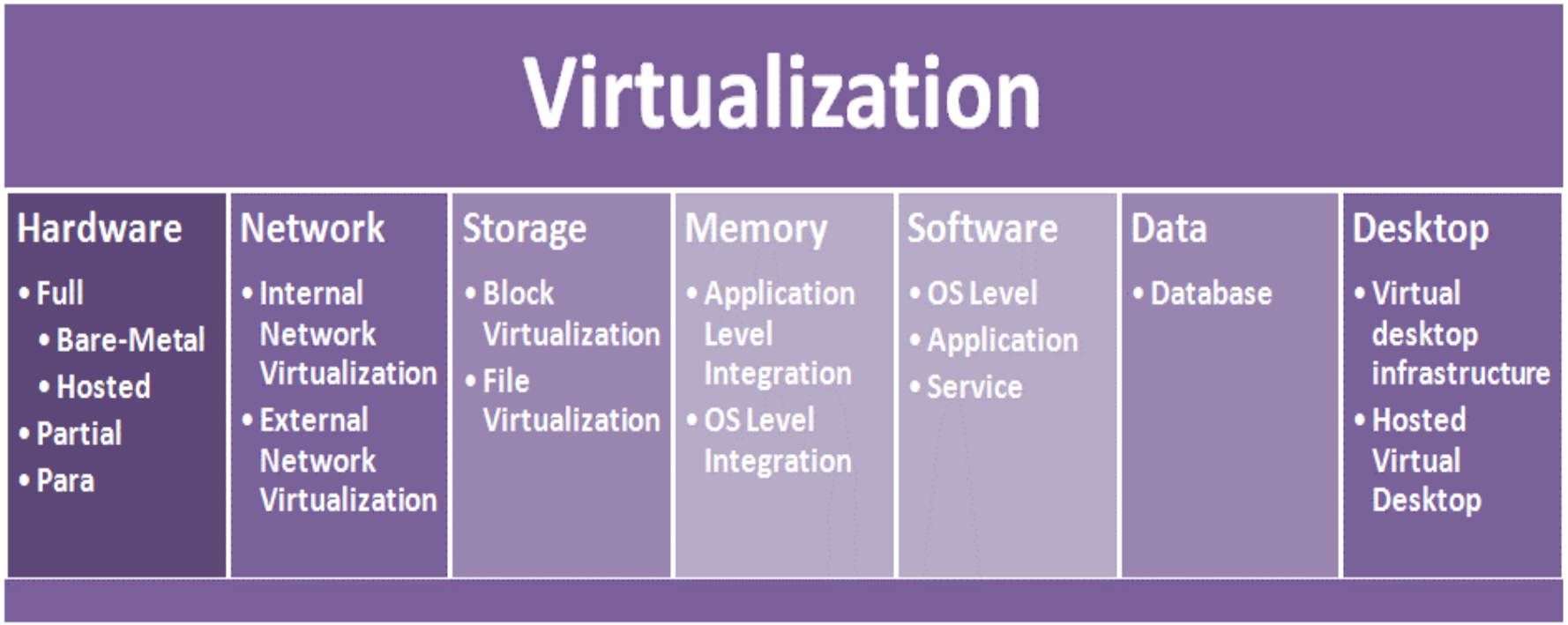
- The software that makes virtualization possible is known as a **Hypervisor**, also known as a **Virtualization Manager**.

- **Hypervisor** sits between the **hardware and the operating system**, and **assigns the amount of access that the applications** and **operating systems have with the processor** and **other hardware resources**. (Hypervisor assigns the CPU and other hardware resources needed to run OS & app)



# TYPES OF VIRTUALIZATION

- Hardware/Server Virtualization
- Network Virtualization
- Storage Virtualization
- Memory Virtualization
- Software Virtualization
- Data Virtualization
- Desktop virtualization



# PROS AND CONS OF VIRTUALIZATION

**Benefits of virtualization offered up by both organizations as well as vendors:**

- **Lower overall capital expenditures.** By means of **virtualization we don't need to purchase a server for every single application in our organization. By hosting multiple virtual servers on a single physical machine, dramatically reduces cost overhead.**

- **Automated tasks**. Virtualization automates a number of significant routine IT tasks. (Automatically starting up of VM, if overload automatically migrating to other server, Automatically backup data, Scaling out & scaling in resources based on load on VM).

- **Greater redundancy. Virtualization should improve our uptime.** Virtualization technologies allow **greater safety and security while reducing the points of contact.** (Virtualization allows to maintain redundant copy, if VM crash or fail over automatically starts up redundant VM)

- **Faster deployment. Deploying a virtual machine is overwhelmingly simpler and quick than deploying a physical machine.**

# PROS AND CONS OF VIRTUALIZATION

**There can be some downsides to virtualization, as well:**

- **High upfront expenditures. When we are planning to implementing a virtualization strategy from the ground up, we need to sink more money into hardware for an immediate future.** While will save in the long run, implementation can get pricey. (A 64-bit processor VM Monitor Mode extensions at least 4 GB of RAM, DMA).

- **Not all applications are ready for virtualization. There are still vendors not fully supporting virtualized environments.** (There are still app developing by vendors that cannot run in an virtualized environment)

- **The danger of server sprawl. Because servers are so easy to deploy in a virtualized environment, there's always the danger that new servers will be added even when they're not needed. Instead of the 10 or 20 virtual servers you really need, you might have 30 or 40.** (Simply deploying an VM even when it is not needed may sprawl /slow server)

# PROS AND CONS OF VIRTUALIZATION

## Pros and Cons of Virtualization

**Pros:**

- Sandbox
- Hardware independent
- OS independent
- Fast Recovery
- Live Backup
- Migrate data
- Reduced Hardware
- Run Multiple OS Simultaneously
- Cost savings
- Use of Multicore processors
- System Security
- Test and Development

**Cons:**

- Less Efficient
- Unstable Performance
- Tools lack ability
- Rapid Deployment
- Latency of Virtual Disk
- Backup and Data Sets
- Security Issues
- Hardware compatibility issues
- Managing and Securing is difficult